汽车电子与控制
Automotive Electronics and Control Systems (AECS)

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汽车安全与节能国家重点实验室

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第二章：汽车电控系统数字核心（单片机）
Digital Core of AECS (Microcontroller)

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1 TC系列MCU结构与编程

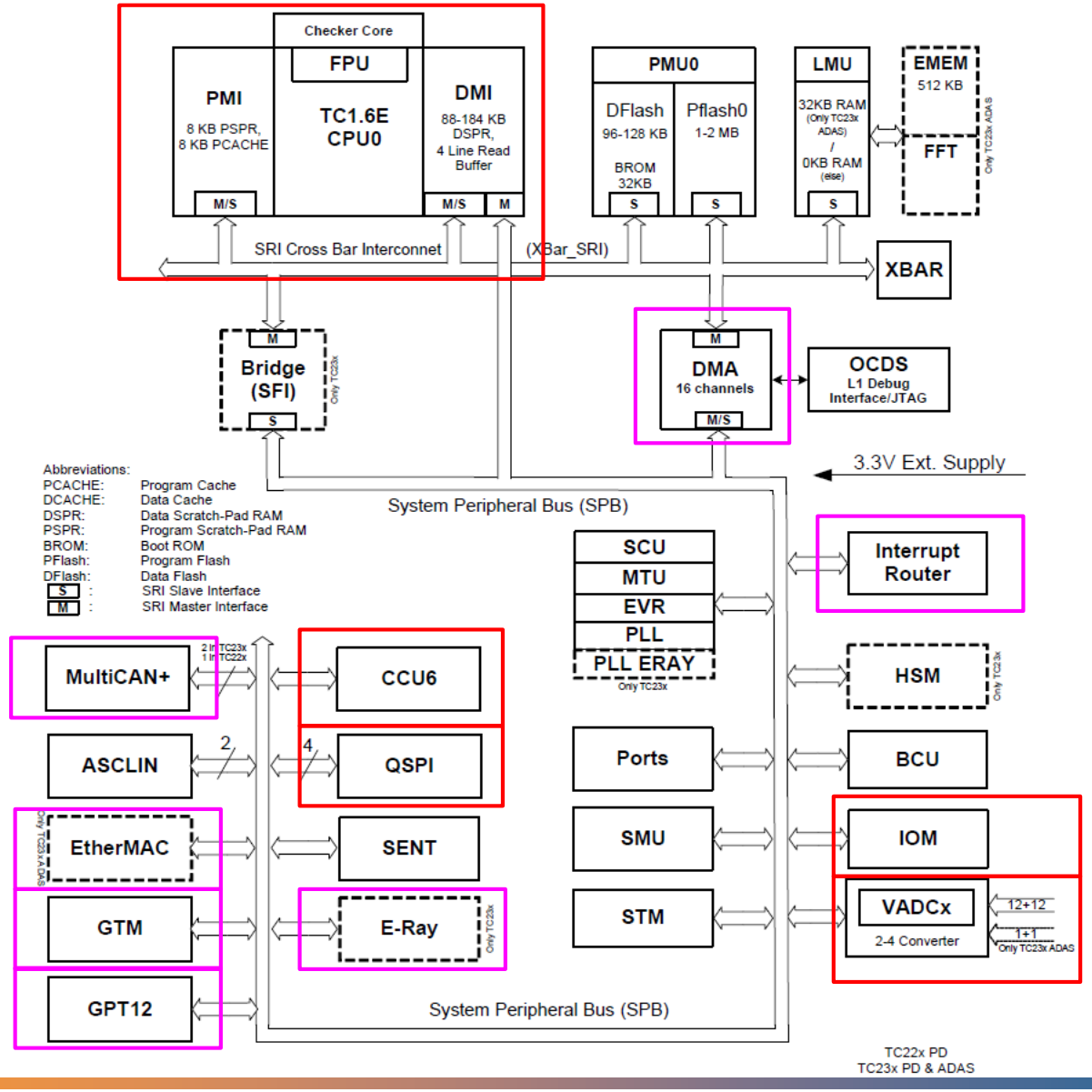
2 TC系列MCU指令系统

3 TC系列MCU的CPU与外围电路

4 TC系列MCU的周边模块

本科阶段应掌握的内容

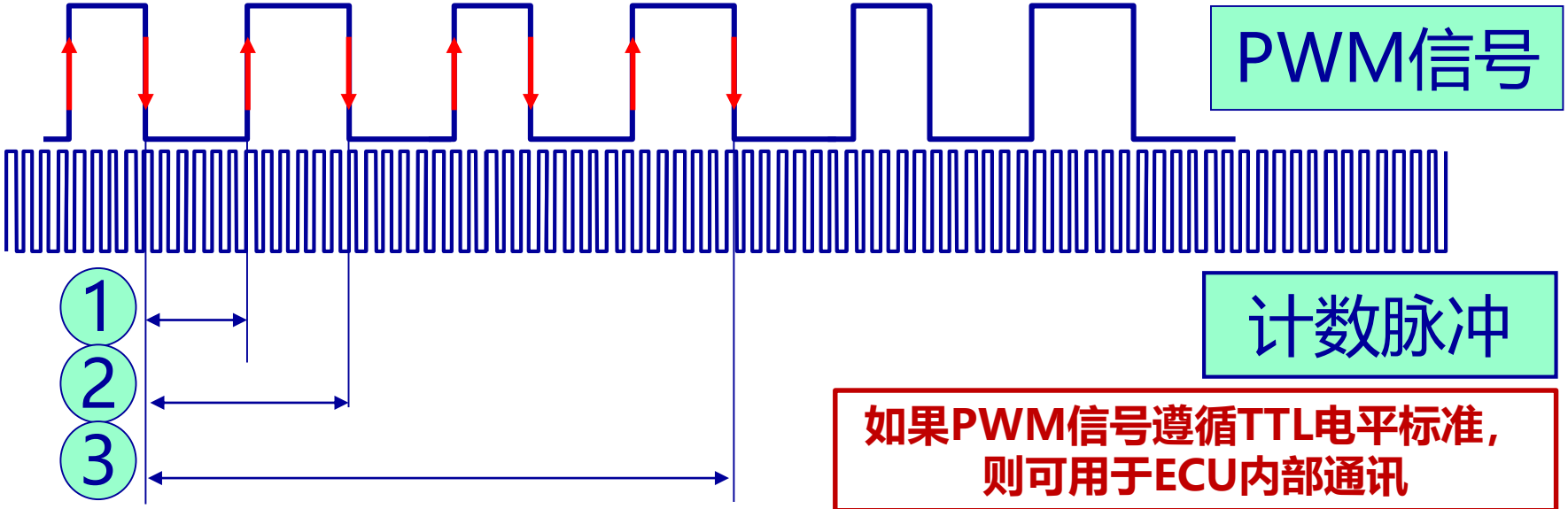
研究生阶段应掌握的内容



- 4.2 CCU6
- 4.3 QSPI

基本概念/ Basic concepts

- PWM信号：脉冲宽度调制（Pulse width modulation，PWM），上升沿，下降沿，占空比/ Rising edge, Falling edge, Duty
- TTL电平：Transistor-Transistor Logic, 晶体管-晶体管逻辑电平，5V->1, 0V->0，计算机/嵌入式控制 设备内部数据传输
- 计数定时器: Counting Timer / Frequency
- 输入捕捉Input Capture、输出比较Output Compare

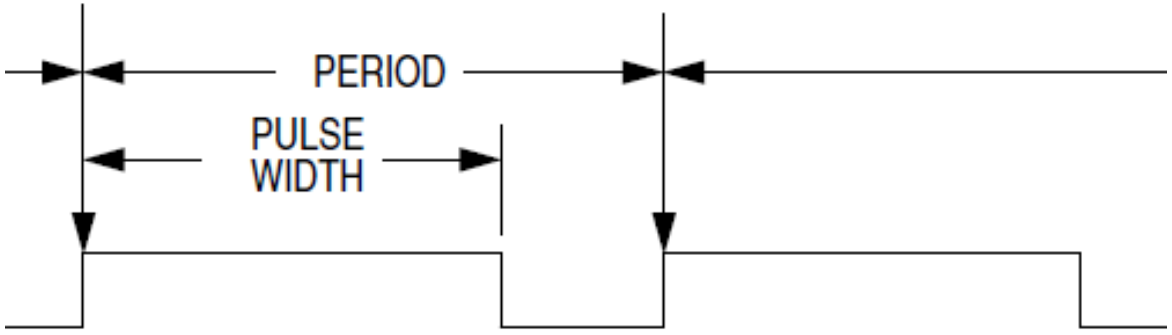


CCU6（Capture and Compare Unit)

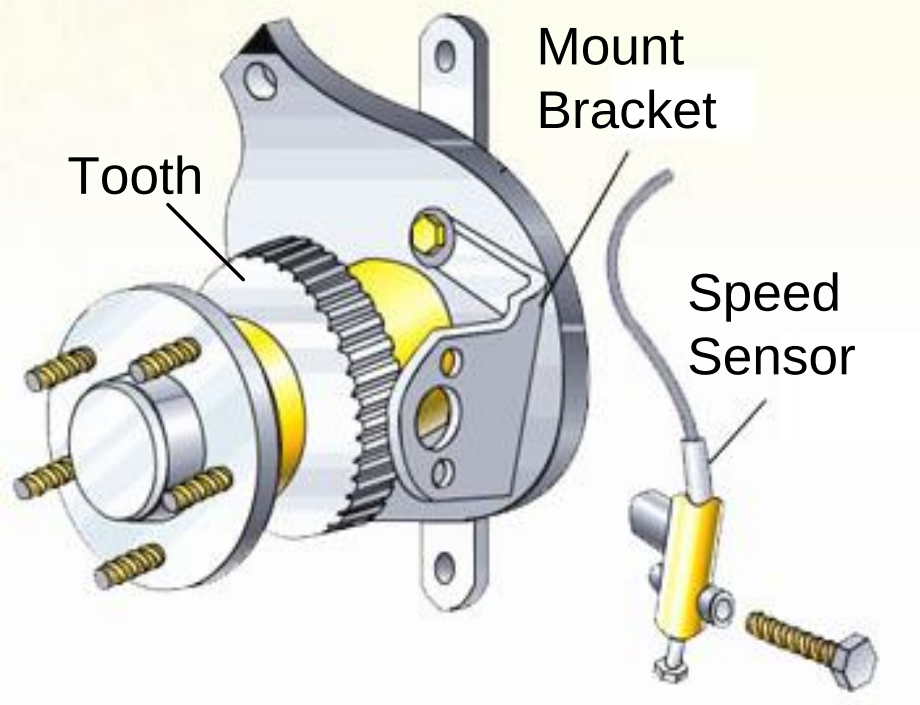
- PWM（Pulse Width Modulation）
- 理论基础
  - 采样控制理论：冲量相等而形状不同的窄脉冲加在具有惯性的环节上时，其效果基本相同。
- 实现：
  - 上世纪80年代之后，随着电力电子器件的发展才得以实现。
  - 对半导体开关器件的导通和关断进行控制，使输出端得到一系列幅值相等而宽度不等的脉冲。
  - 配合适当的RC或LC电路进行滤波

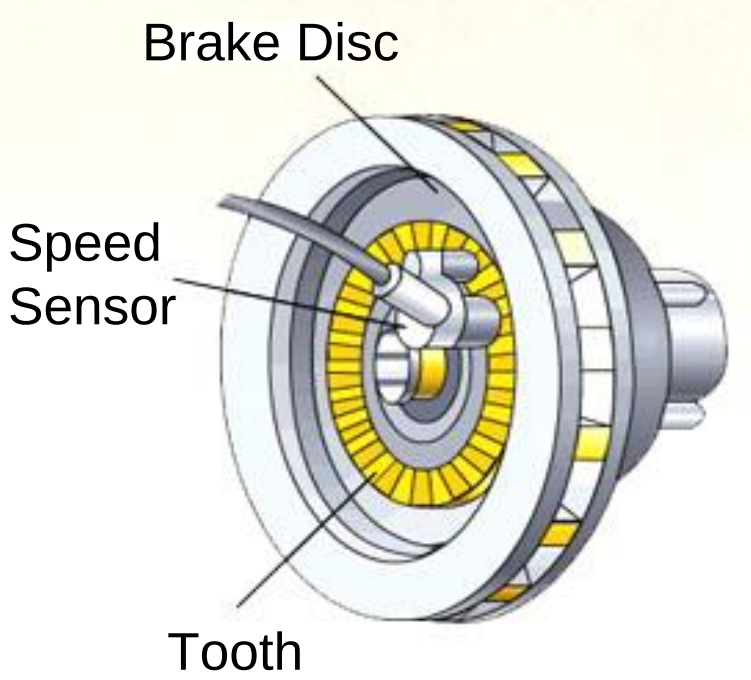
CCU6（Capture and Compare Unit)

- 优势：
  - 利用微处理器的数字输出对模拟电路进行控制，无需进行模数转换。
  - 对噪声抵抗能力相对于模拟控制更强。
- 主要参数
  - 周期：Period
  - 占空比： Duty Ratio=Pulse Width/Period
    高电平在一个周期内所占的时间比例



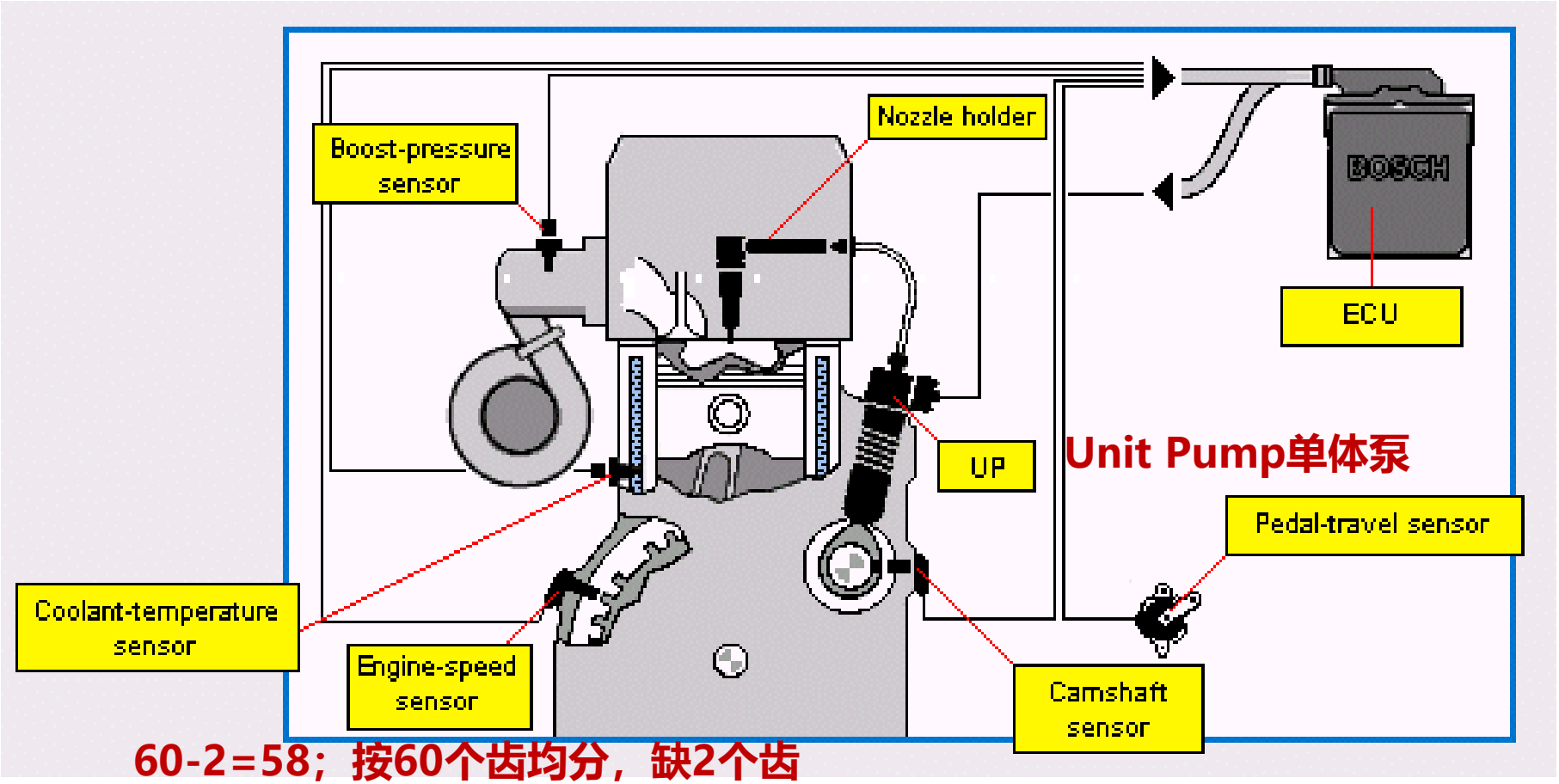
- **典型采用输入捕捉的场合:ABS系统/ Typical Application for IC and OC: ABS**





CCU6（Capture and Compare Unit)

- **典型采用输入捕捉的场合:喷射控制/ Typical application: Injection Control for Engines**



CCU6（Capture and Compare Unit）

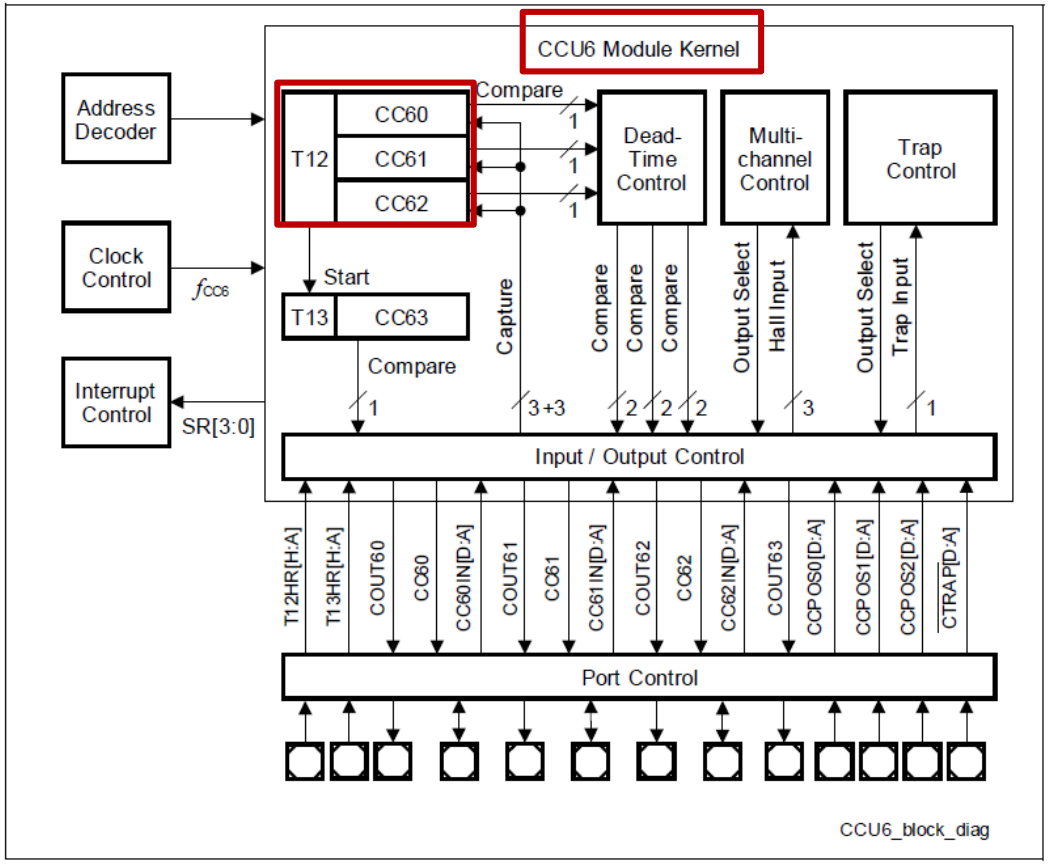
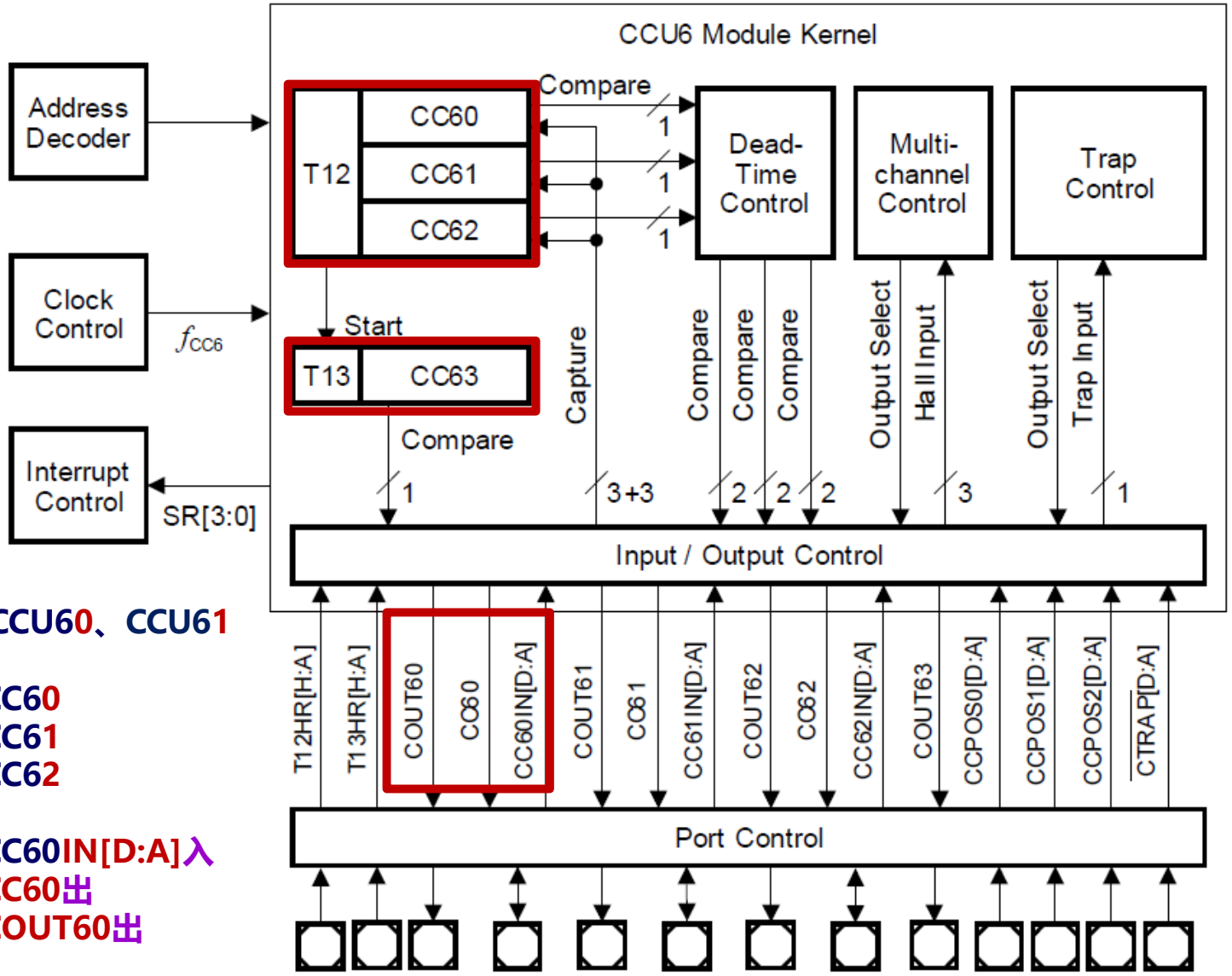


Figure 25-1 CCU6 Block Diagram

- **T12 有三个通道，可以用于输入捕捉和输出比较**
- **T13有一个通道，仅可以用于输出比较**

- **输入捕捉目的：获取两个事件之间的时间差**
- **输出比较目的：以一定频率、占空比输出方波信号，即输出PWM信号**
- **PWM：脉宽调制（Pulse-Width Modulation），是一种常见的反馈和控制信号**
- **事件：上升沿、下降沿**

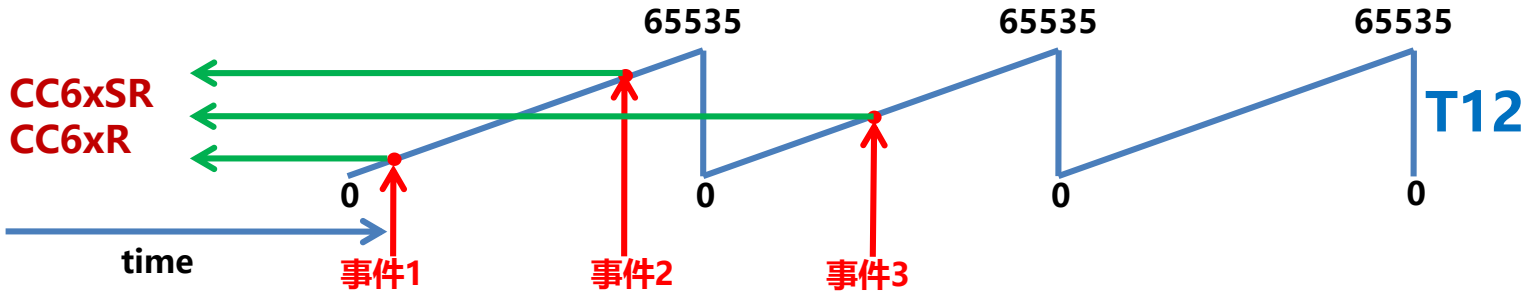
CCU6 (Capture and Compare Unit)

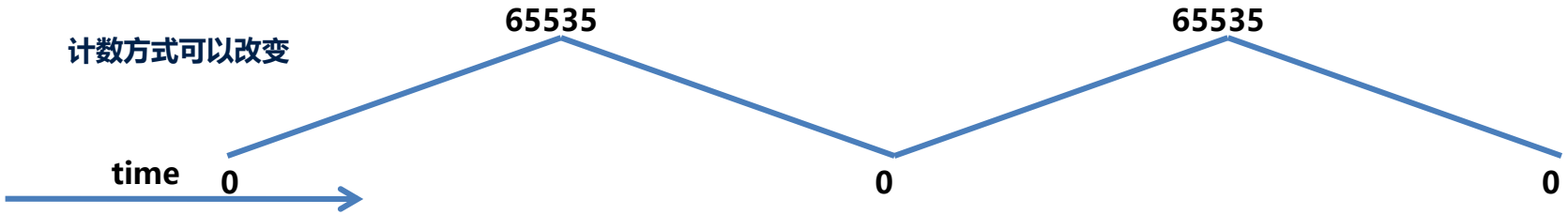


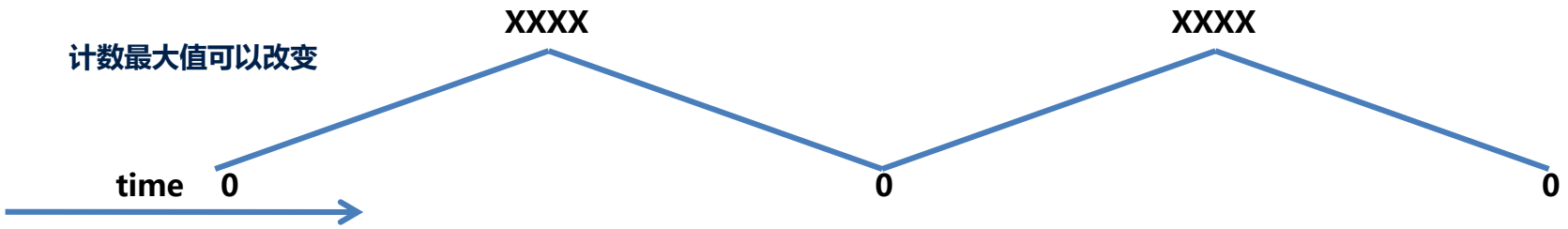
第一层：CCU60、CCU61
第二层：
CCU60_CC60
CCU60_CC61
CCU60_CC62
第三层：
CCU60_CC60IN[D:A]入
CCU60_CC60出
CCU60_COUT60出

CCU6（Capture and Compare Unit)

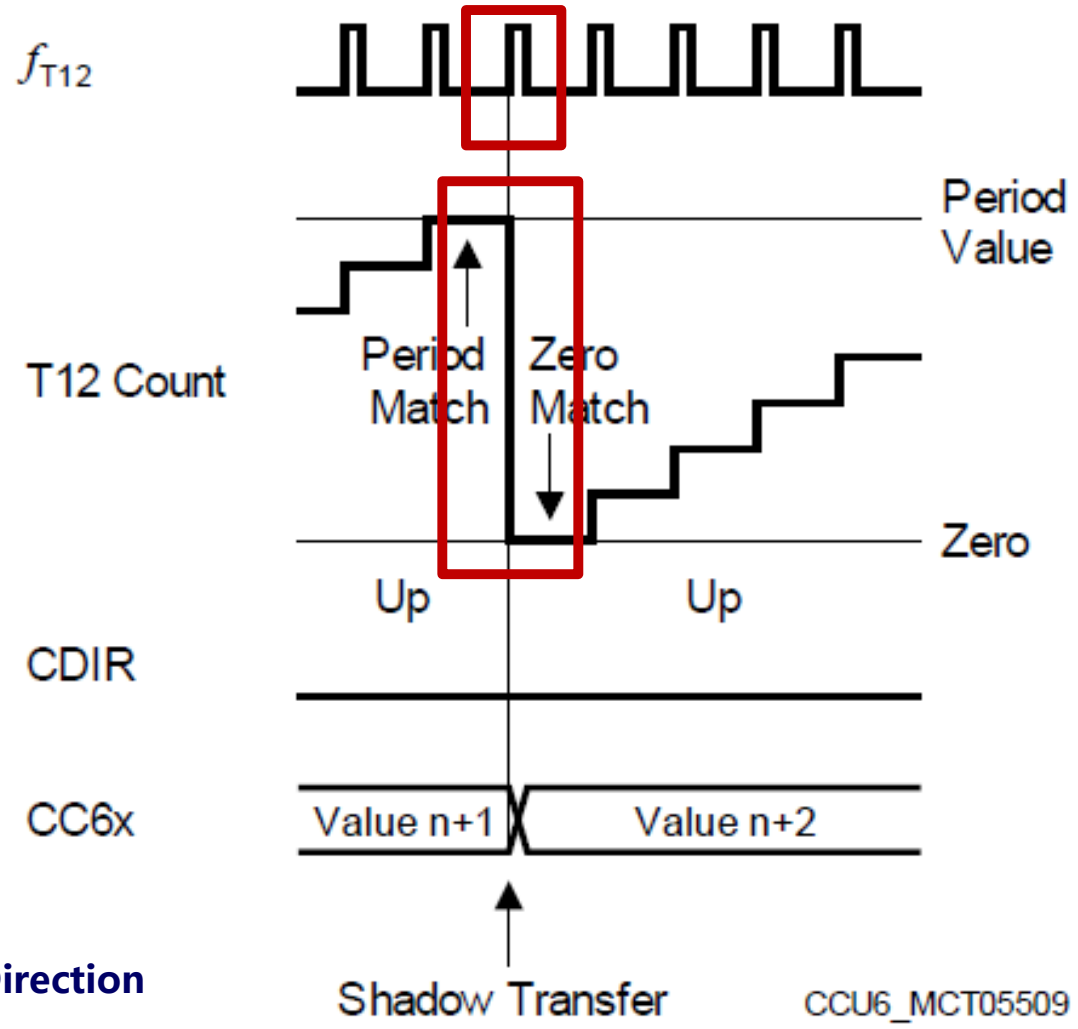
- **输入捕捉原理：**
  - **捕捉事件（Rising Edge/Falling Edge/Both），记录对应的计数寄存器的值，从而计算事件之间的时间差**





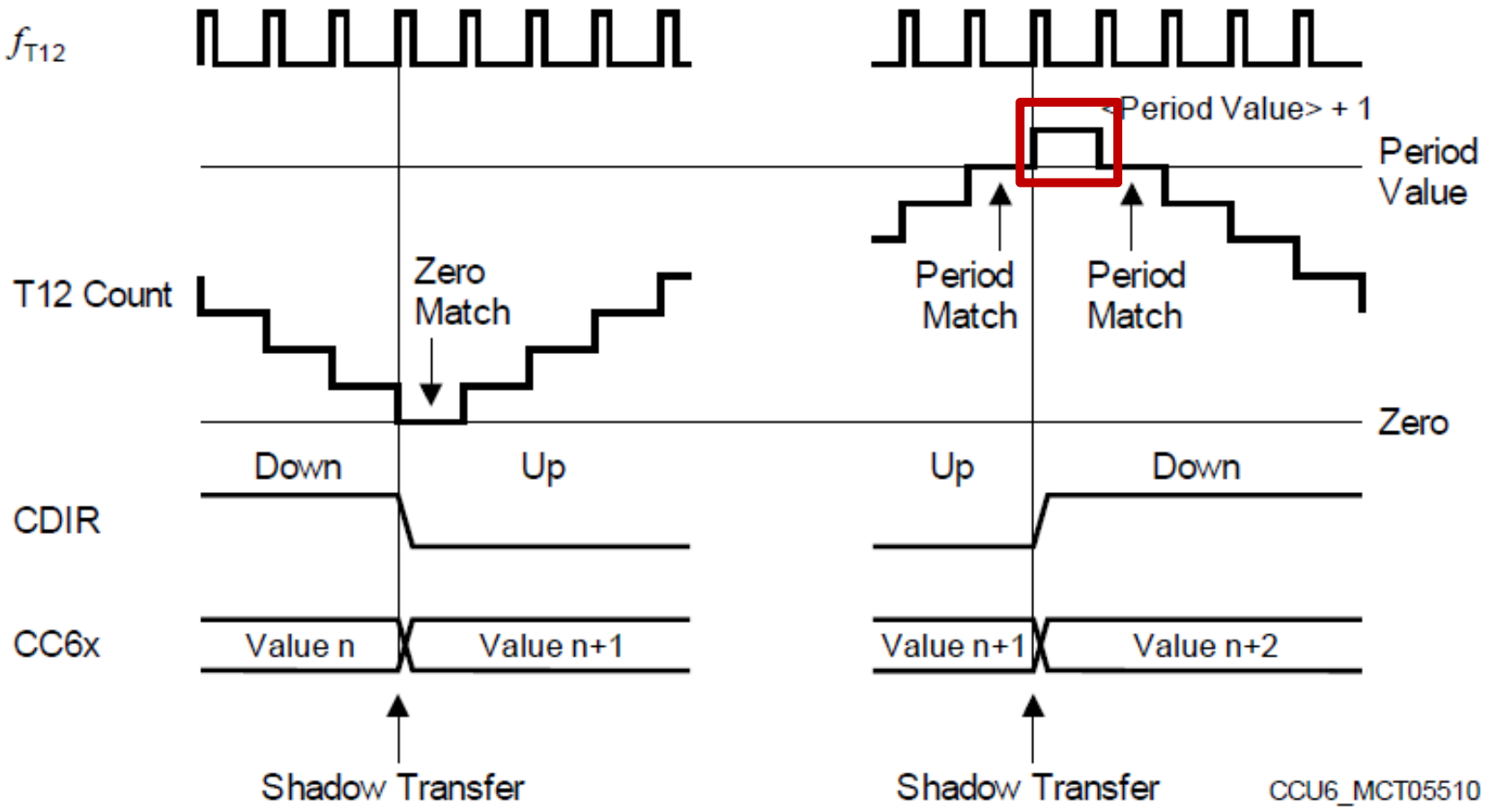


计数方式1
Edge Aligned
边沿对齐



CDIR: Current Counting Direction

计数方式2 Center Aligned 中心对齐



CCU6的输入捕捉模式设定：T12MSEL寄存器

Field	Bits	Type	Description
MSEL60, MSEL61, MSEL62	[3:0], [7:4], [11:8]	rw	**Capture/Compare Mode Selection** These bit fields select the operating mode of the three T12 capture/compare channels. Each channel (x = 0, 1, 2) can be programmed individually for one of these modes (except for Hall Sensor Mode). Coding see Table 25-5.
HSYNC	[14:12]	rw	**Hall Synchronization** Bit field HSYNC defines the source for the sampling of the Hall input pattern and the comparison to the current and the expected Hall pattern bit fields. Coding see Table 25-11.
DBYP	15	rw	**Delay Bypass** DBYP controls whether the source signal for the sampling of the Hall input pattern (selected by HSYNC) is delayed by the Dead-Time Counter 0. 0_B The bypass is not active. Dead-Time Counter 0 is generating a delay after the source signal becomes active. 1_B The bypass is active. Dead-Time Counter 0 is not used for a delay.
0	[31:16]	r	**Reserved;** Returns 0 if read; should be written with 0.

T12MSEL
T12 Mode Select Register **(68_H)** **Reset Value: $0000\ 0000_H$**

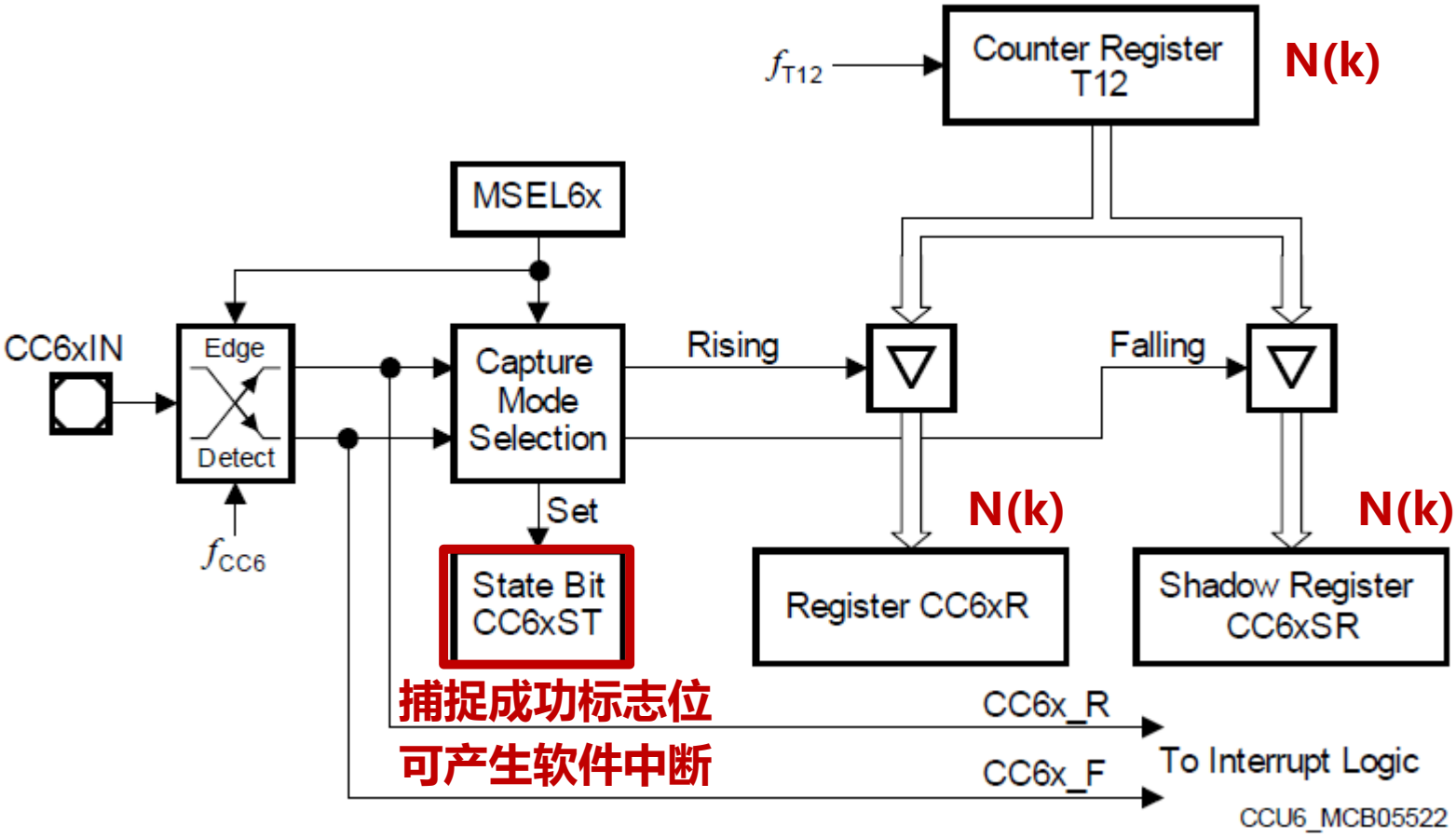
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0															
r															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
D BYP	HSYNC			MSEL62				MSEL61				MSEL60			
rw	rw			rw				rw				rw			

Table 25-3 Capture Modes Overview

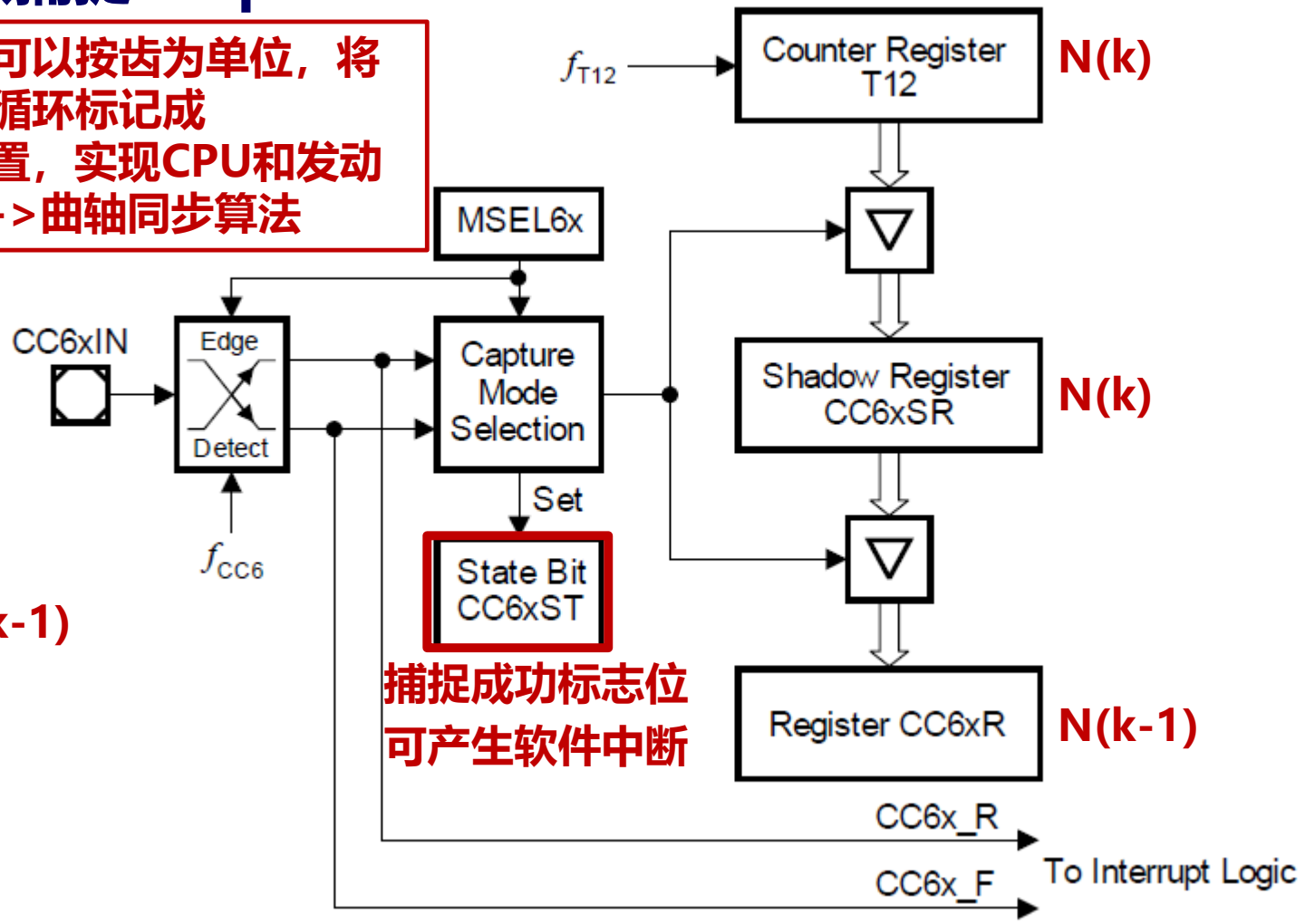
MSEL6x	Mode	Signal	Active Edge	CC6nSR Stored in	T12 Stored in
0100_B	1	CC6xIN	Rising	–	CC6xR
		CC6xIN	Falling	–	CC6xSR
0101_B	2	CC6xIN	Rising	CC6xR	CC6xSR
0110_B	3	CC6xIN	Falling	CC6xR	CC6xSR
0111_B	4	CC6xIN	Any	CC6xR	CC6xSR

CCU6的输入捕捉: Capture Mode 1



CCU6的输入捕捉: Capture Mode 2~4

通过这些技术，可以按齿为单位，将发动机一个工作循环标记成58*2=116个位置，实现CPU和发动机工作状态同步->曲轴同步算法



捕捉成功标志位
可产生软件中断

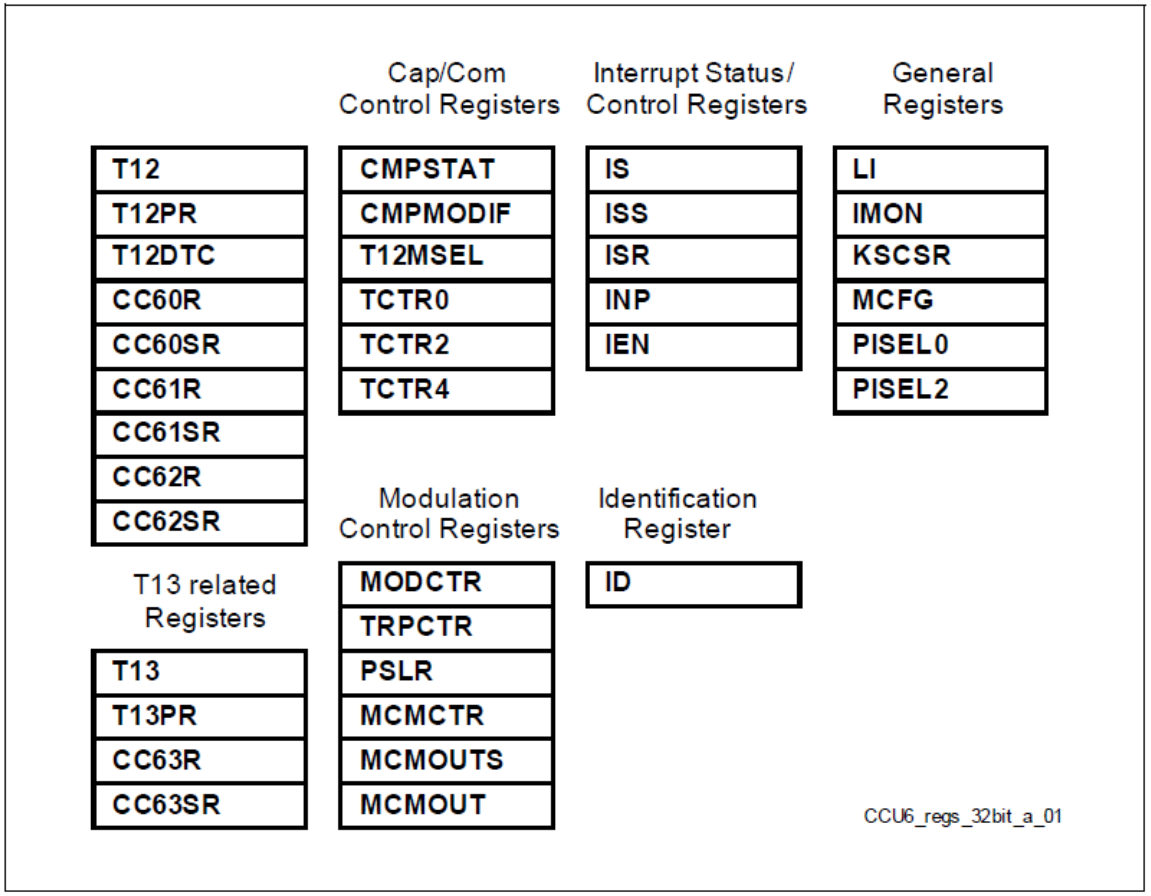
$\delta N = N(k) - N(k-1)$

Rising edge-2

Falling edge-3

Both edge-4

相关寄存器（具体参考实验）



- 本实验用到**输入捕捉功能**
- 需要配置**CCU6输入捕捉相关寄存器、单片机端口的IOCR寄存器（参考第一次实验）**
- 采用中断需要配置**中断相关寄存器（已经配置好，供大家学习）**

CCU6（Capture and Compare Unit)

```
int core0_main(void)
{
   //设置fCPU=100MHz，fSPB=100MHz//
   IfxScuCcu_Config scuccuconfig;
   IfxScuCcu_initConfig(&scuccuconfig);
   IfxScuCcu_init(&scuccuconfig);
   IfxCpu_disableInterrupts();      //禁止中断
   IfxScuWdt_disableCpuWatchdog(IfxScuWdt_getCpuWatchdogPassword());
   IfxScuWdt_disableSafetyWatchdog(IfxScuWdt_getSafetyWatchdogPassword());

   CAP_Init();//CCU6模块初始化

   //中断初始化
   volatile Ifx_SRC_SRCR *src;
   src = &(MODULE_SRC.CCU6.CCU6[0].SR0);//um的16章table16.4 1320/3435页
   IfxSrc_init(src, 0, ISR_RISE_EDGE);
   IfxSrc_enable(src);
   src = &(MODULE_SRC.CCU6.CCU6[0].SR1);
   IfxSrc_init(src, 0, ISR_TIMEOUT);
   IfxSrc_enable(src);

   IfxCpu_enableInterrupts();     //允许中断

   while (1)
   {
      ;
   }
   return 0;
}
```

主程序：

- 禁止中断
- 初始化时钟
- 配置看门狗
- CCU6模块初始化
- 中断初始化
- 死循环

CCU6（Capture and Compare Unit)

```
#include "SysSe/Bsp/Bsp.h"
#include "IfxScuWdt.h"
#include "Ifx_reg.h"
#include "IfxScuCcu.h"

#define ISR_TIMEOUT  11          //定义中断优先级
#define ISR_RISE_EDGE 10

IFX_INTERRUPT (Timeout, 0, ISR_TIMEOUT)
{
  //溢出中断执行的函数，记录溢出次数，自行编写
}
IFX_INTERRUPT (Rise_edge, 0,ISR_RISE_EDGE)
{
  //上升沿中断执行的函数，计算信号频率，周期，自行编写
}

void CAP_Init(void)
{
   Ifx_CCU6 *cap = &MODULE_CCU60;     //定义指针指向CCU60模块
   uint16 psw = IfxScuWdt_getCpuWatchdogPassword();
   IfxScuWdt_clearCpuEndinit(psw);
   cap->CLC.B.DISR=0; //CCU6模块使能
   while(cap->CLC.B.DISS!=0)          //等待模块开启后再进行后续操作
   {       ;   }
   IfxScuWdt_setCpuEndinit(psw);

  //CCU6输入捕捉相关寄存器配置

  //对应IO口设置为输入
}
```

CCU6输入捕捉程序结构

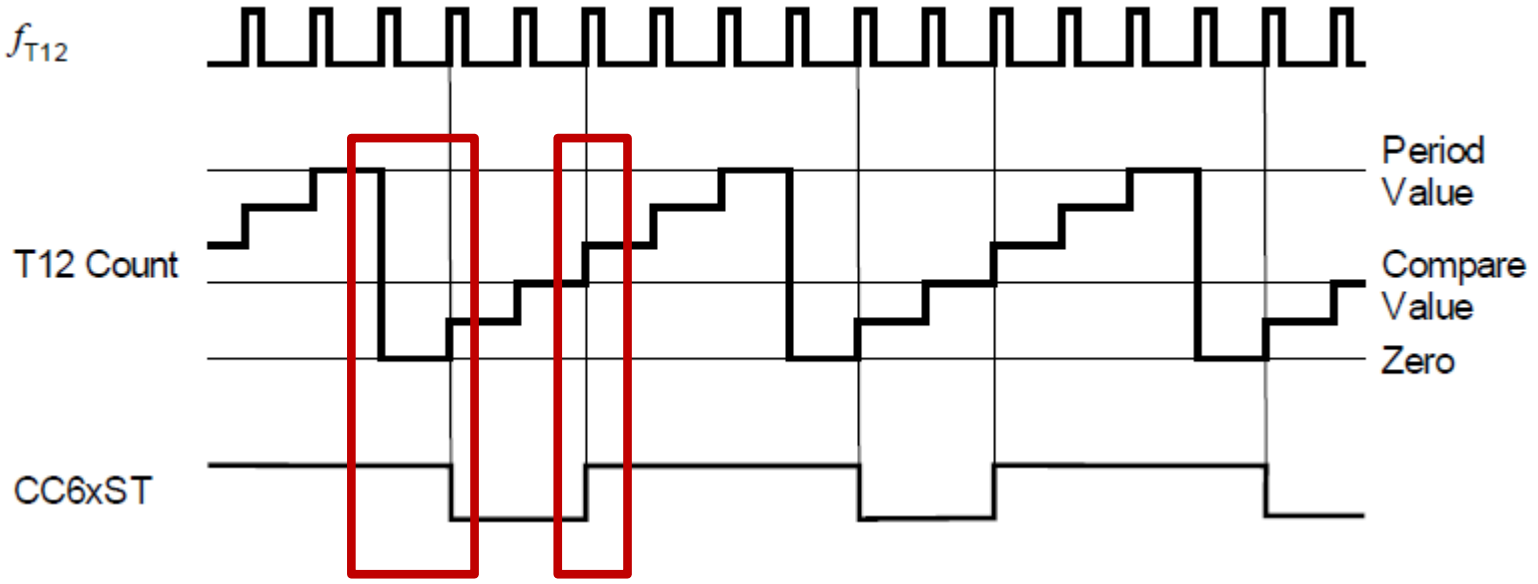
定义中断函数

定义CCU6模块初始化函数：

- **模块使能**
- **输入捕捉设置**
- **配置对应I/O端口为输入**

需要同学们自己编写

Compare Operation, Edge-Aligned Mode（边沿对齐)

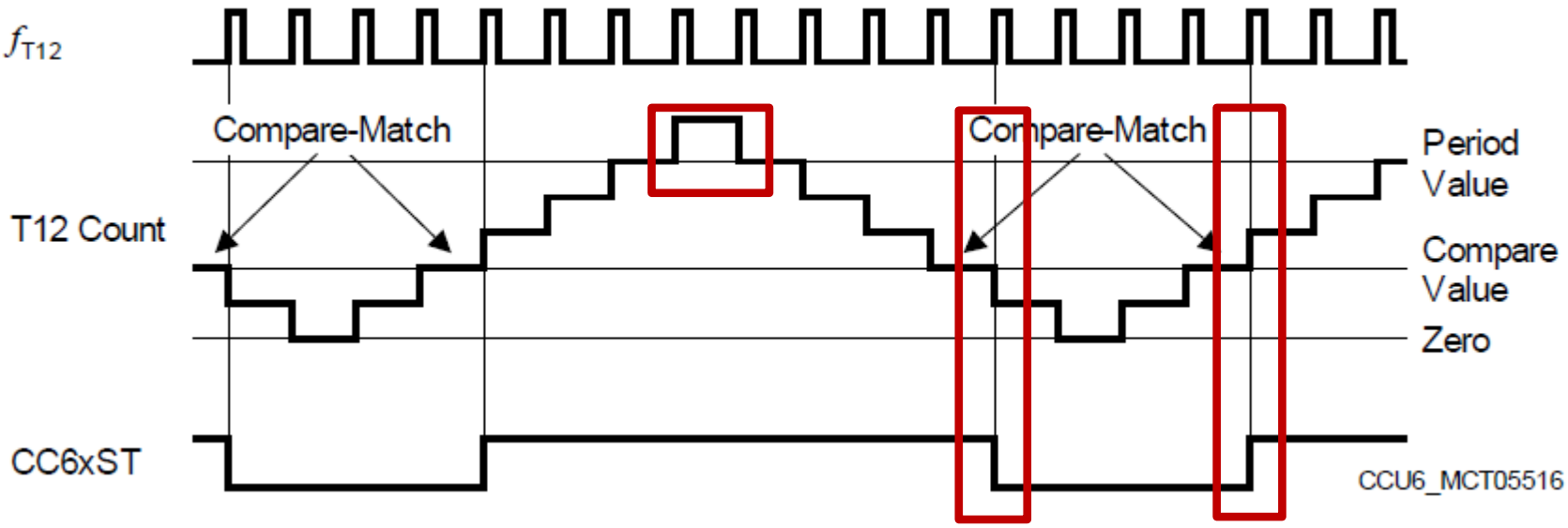


CC6xST代表输出比较的逻辑状态值，该值与真实电平的关系可用寄存器设定

① Zero match: CC6xST维持不变

② T12 Count > Compare value, CC6xST=1; otherwise, =0;

Compare Operation, Center-Aligned Mode （中心对齐)

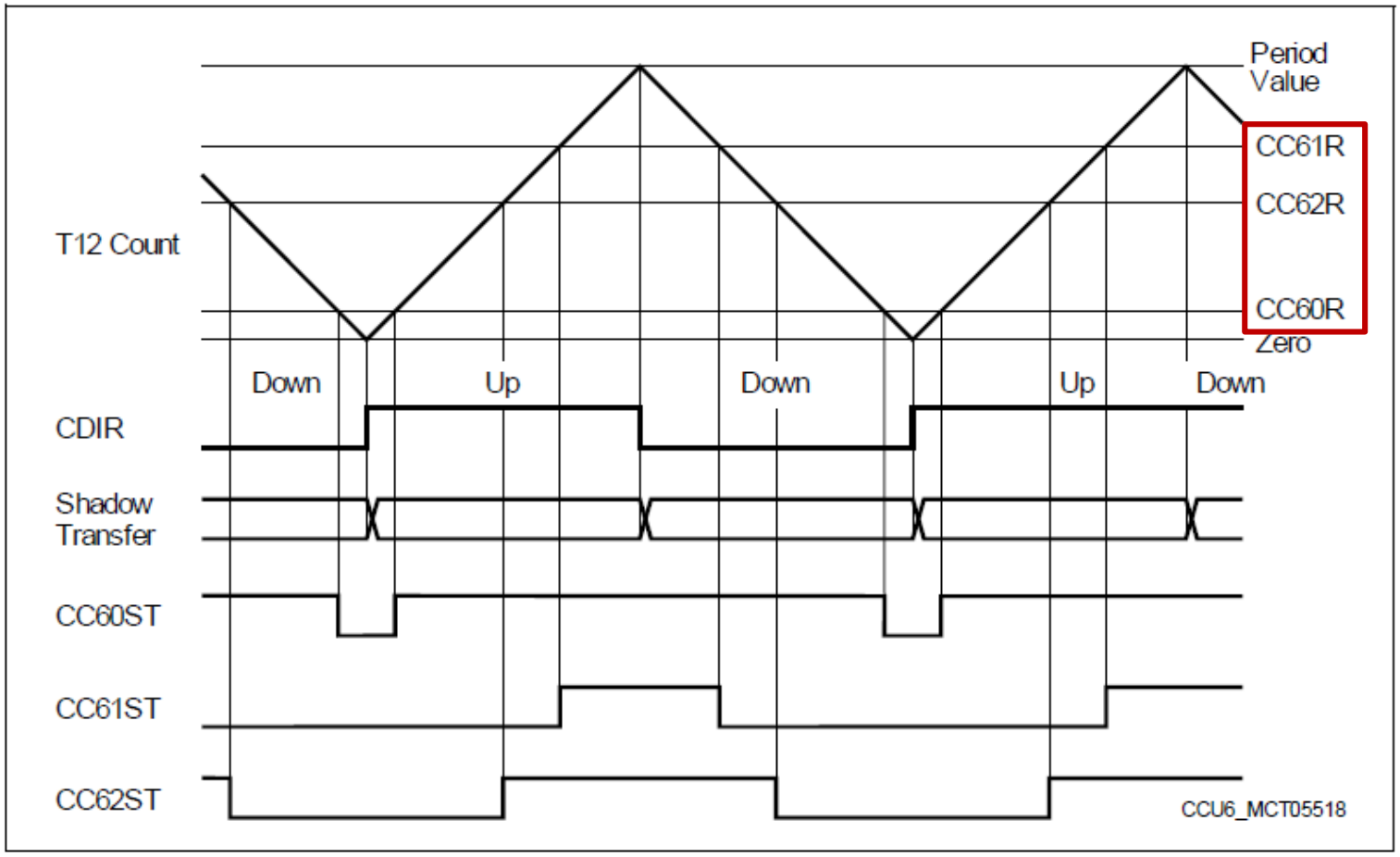


CC6xST代表输出比较的逻辑状态值，该值与真实电平的关系可用寄存器设定

① Compare match: CC6xST维持不变

② T12 Count > Compare value, CC6xST=1; otherwise, =0;

多路PWM模式

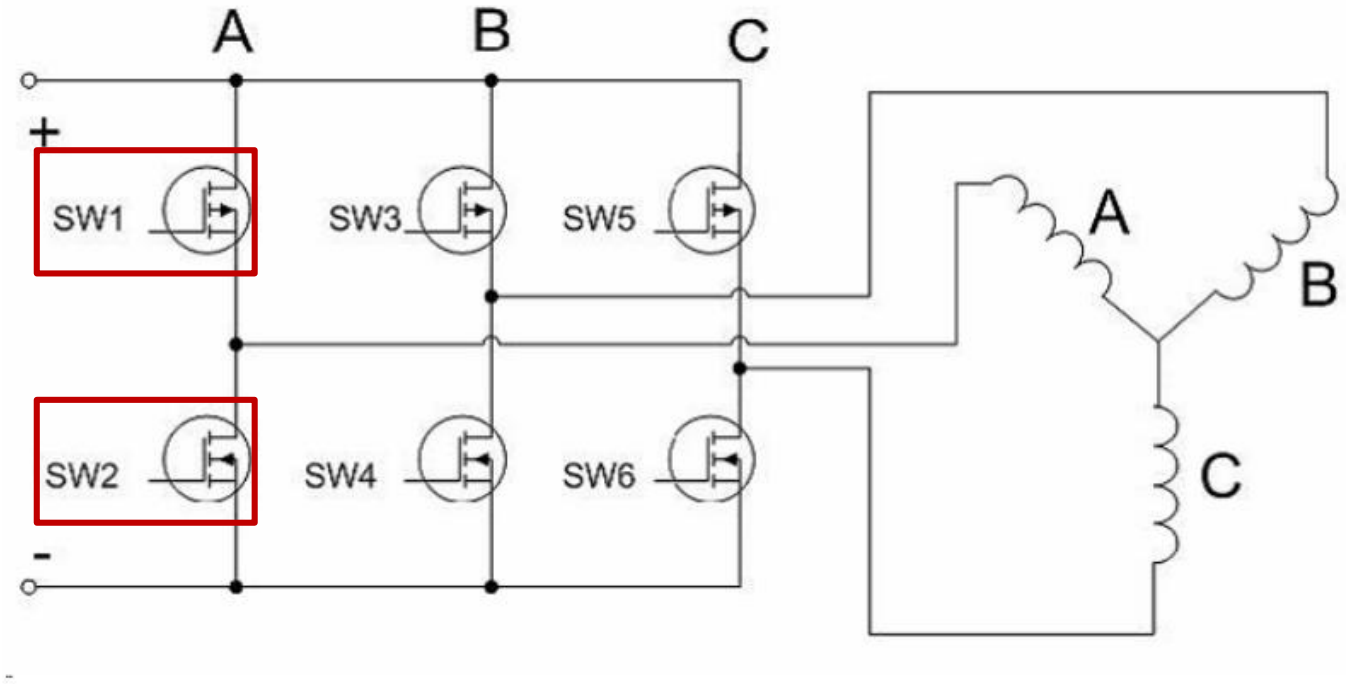


共同的T12计数器
不同的比较阈值

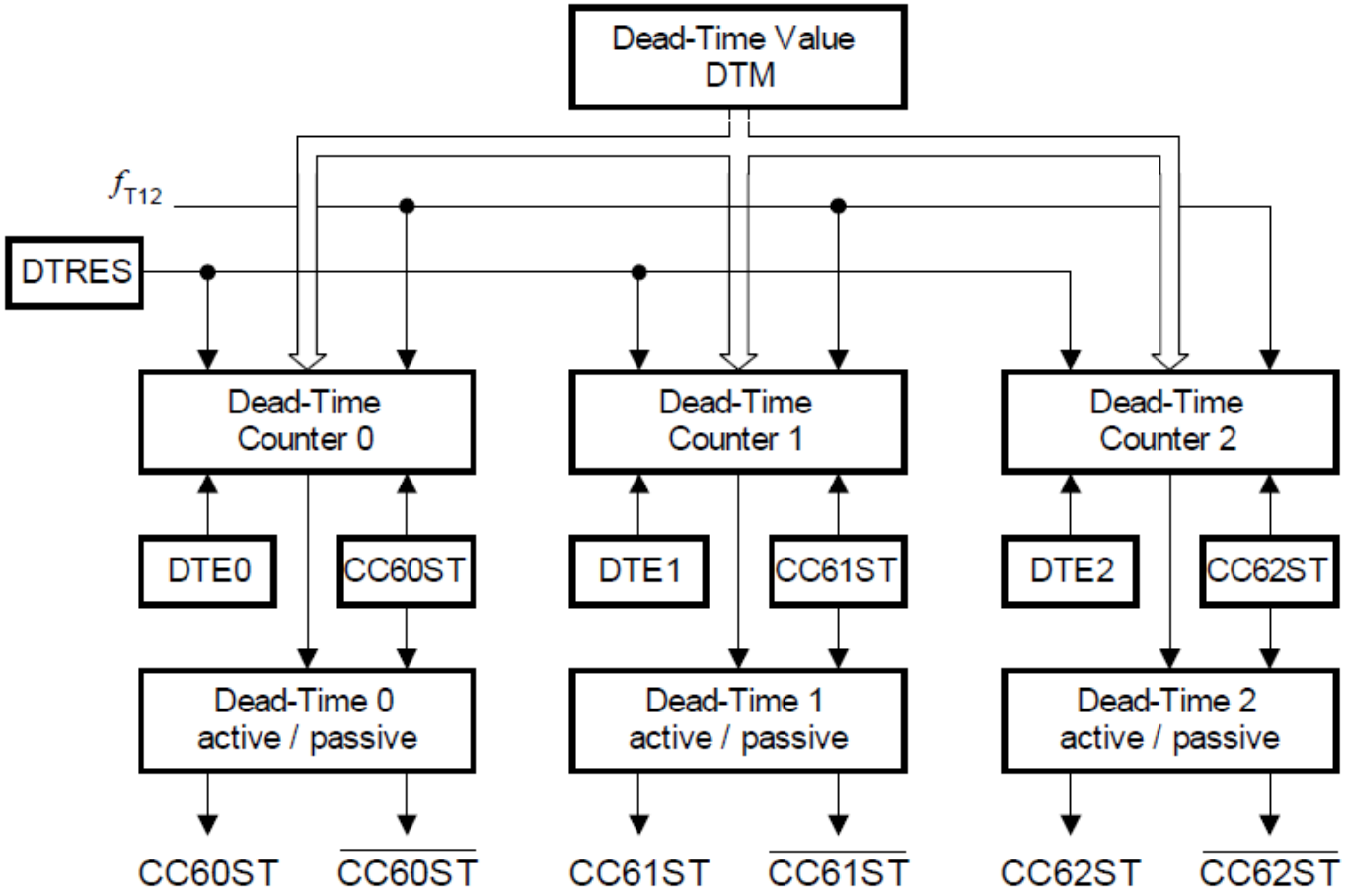
相同的周期，
不同的占空比

Dead Time Compensation （死区补偿)

电机驱动中，用PWM波控制SW1和SW2开关状态切换的时候，需要增加死区防止某一时刻SW1和SW2同时导通

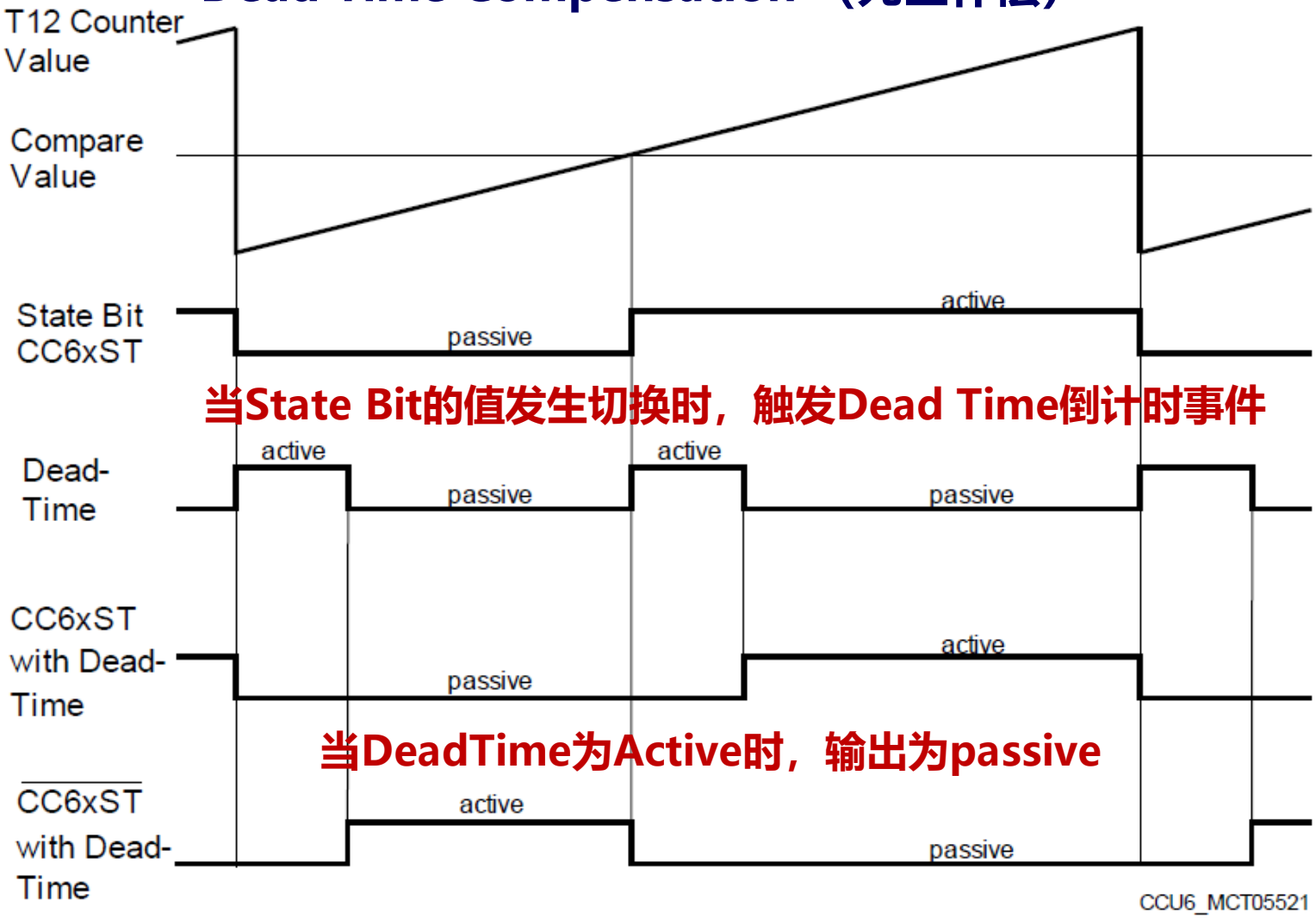


Dead Time Compensation (死区补偿)

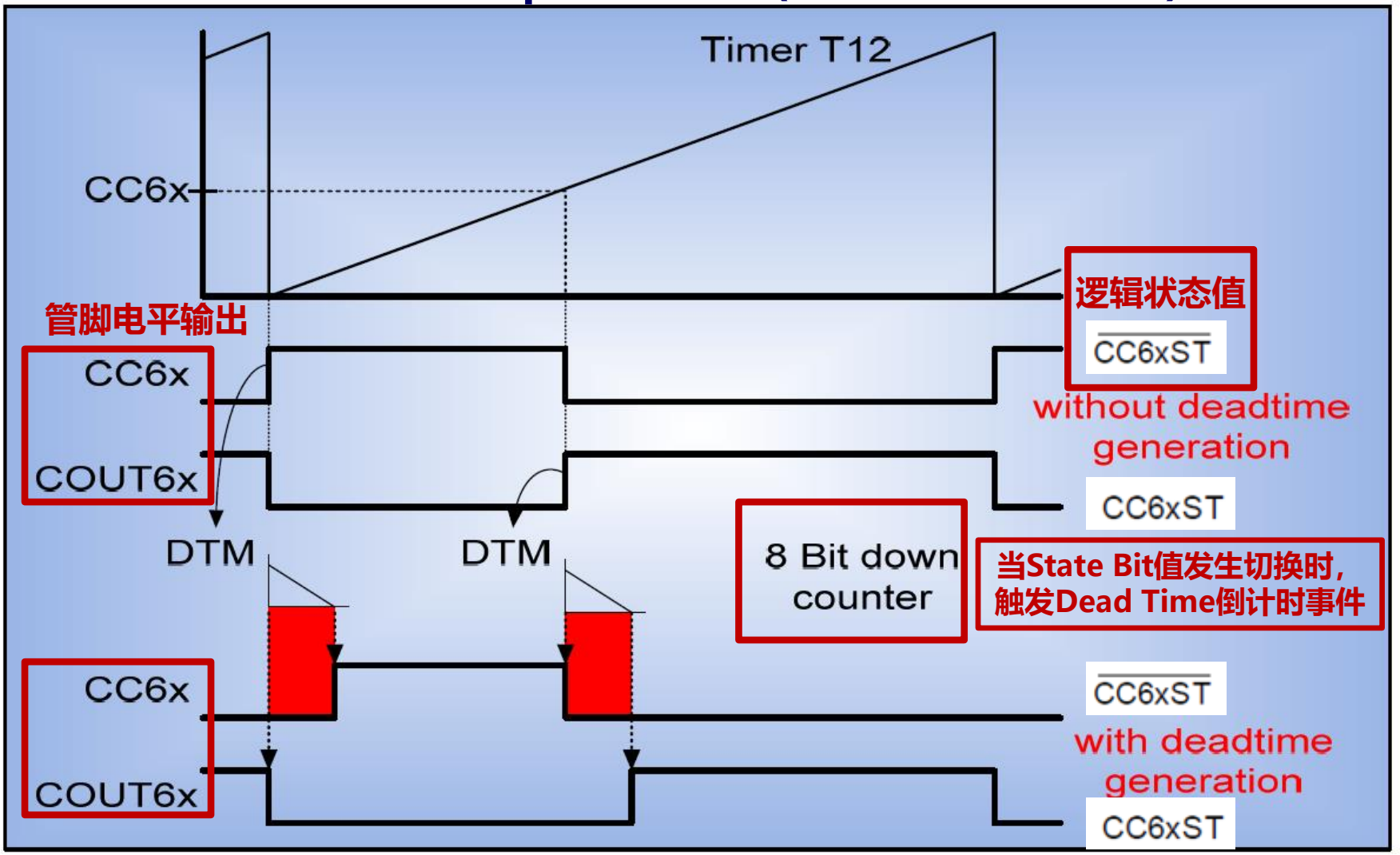


CCU6（Capture and Compare Unit)

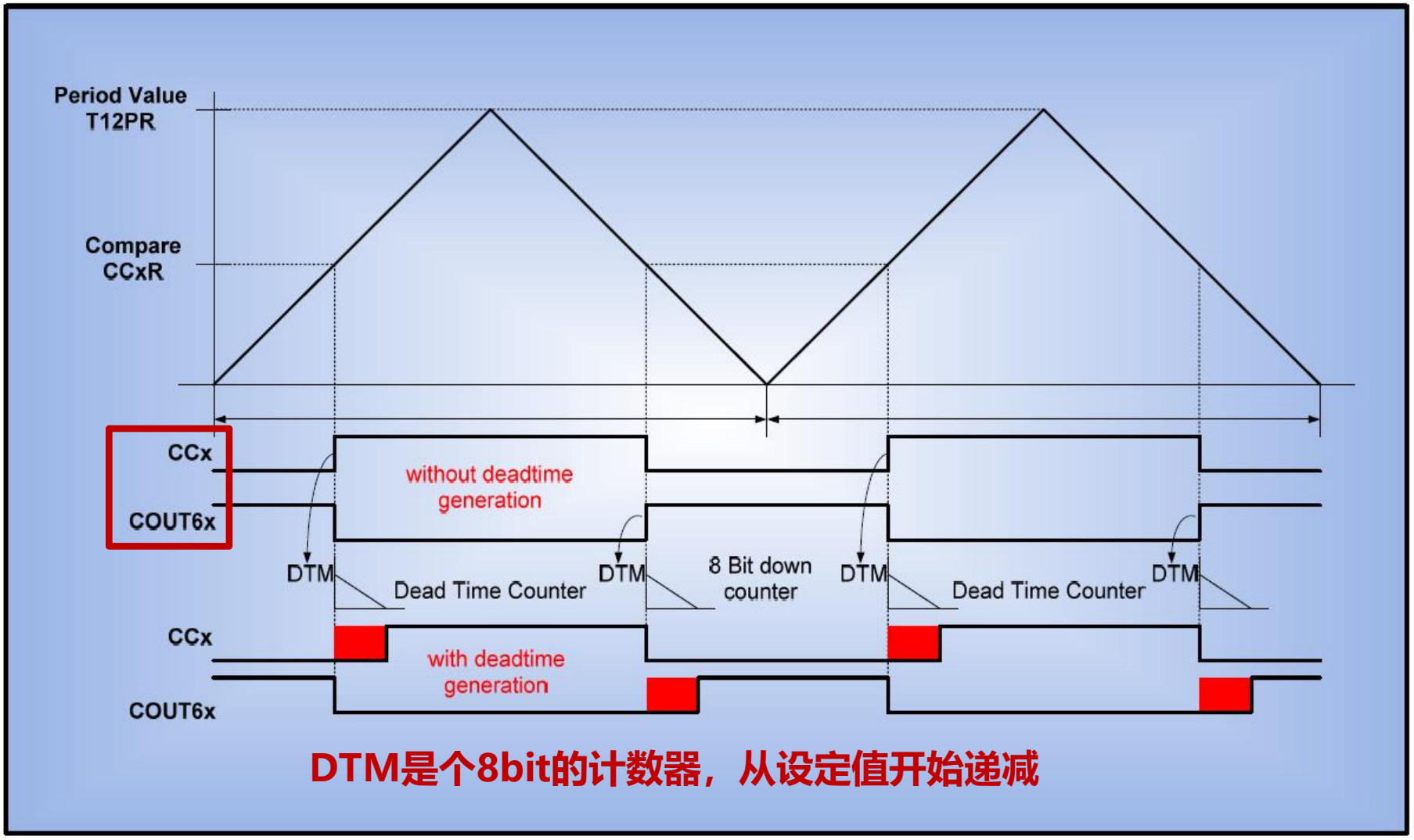
Dead Time Compensation （死区补偿)



Dead Time Compensation (死区补偿+边沿对齐)

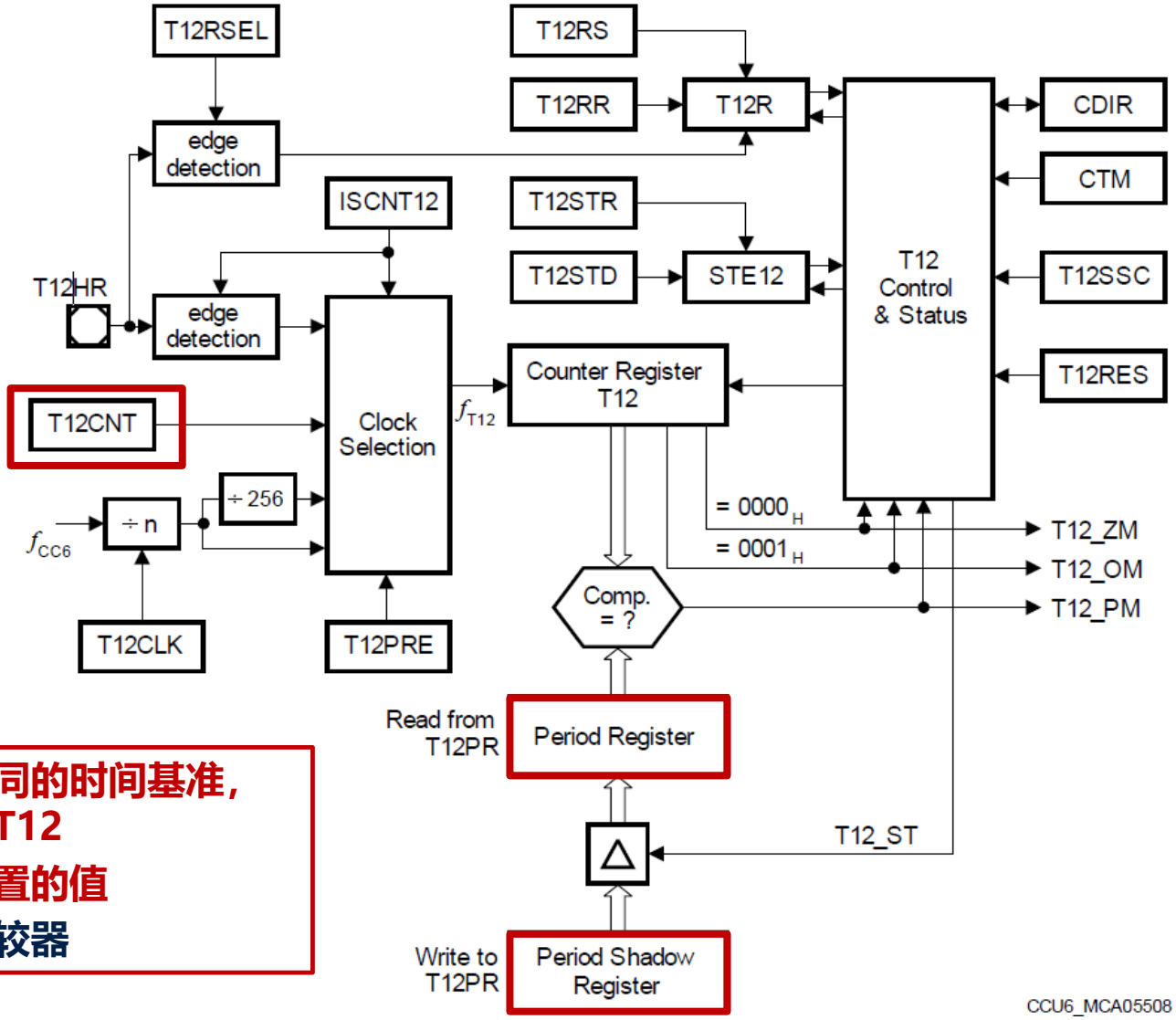


Dead Time Compensation （死区补偿+中心对齐)



DTM是个8bit的计数器，从设定值开始递减

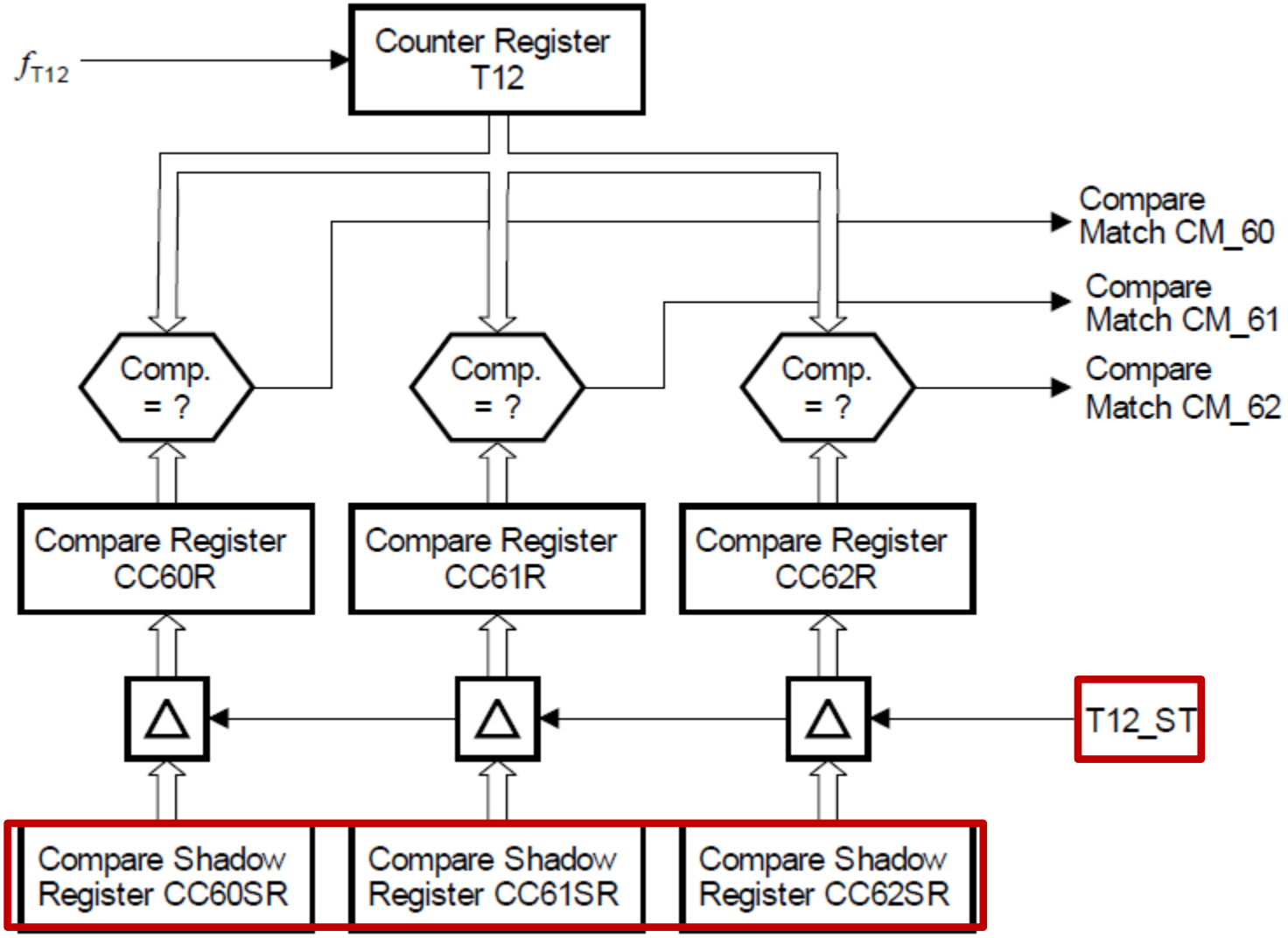
CCU6 (Capture and Compare Unit)



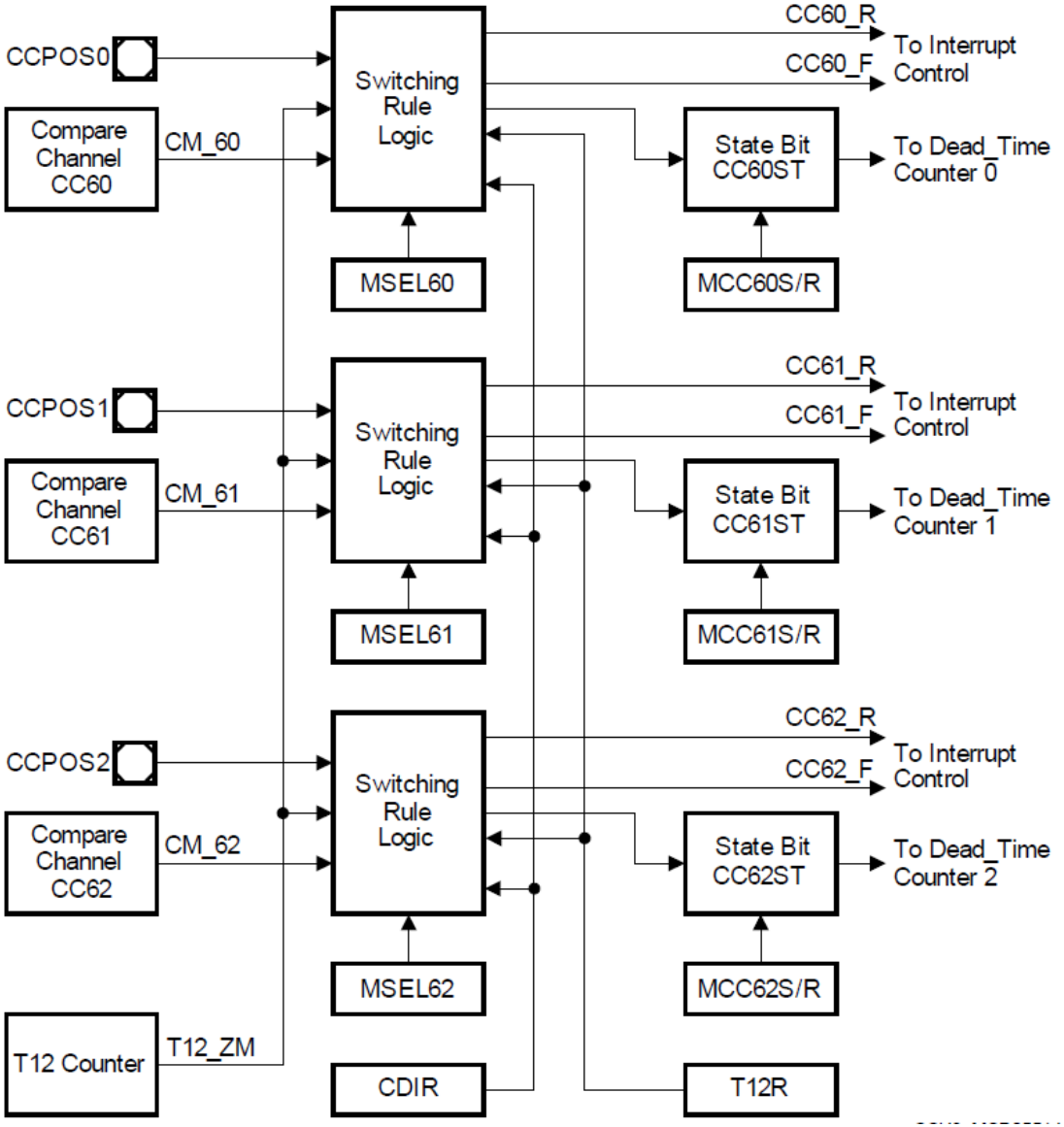
- 共同的时间基准，如T12
- 预置的值
- 比较器

CCU6的输出比较

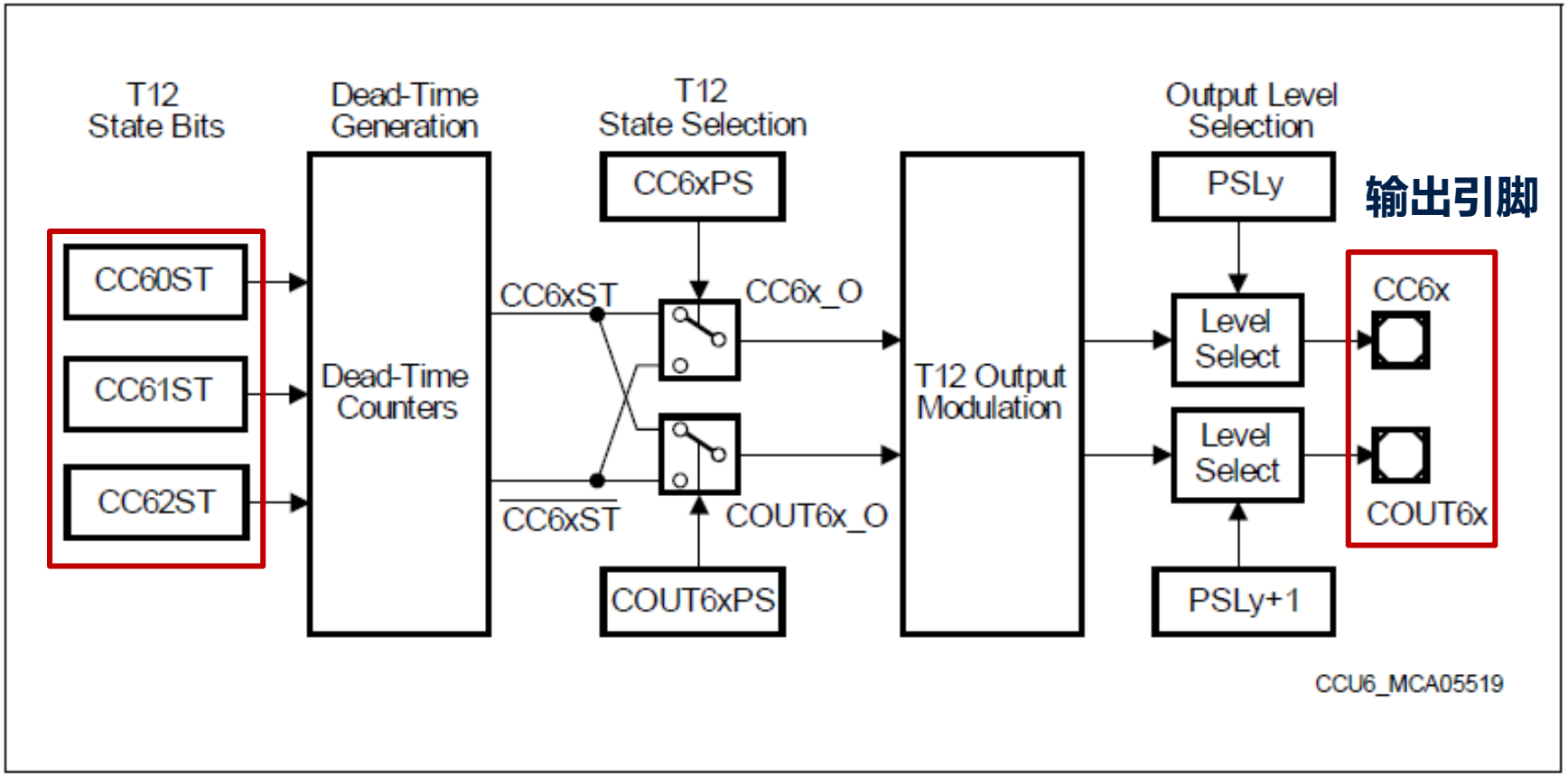
CCU6的输出比较


f_{T12}
Counter Register T12
Compare Match CM_60
Compare Match CM_61
Compare Match CM_62
Comp. = ?
Comp. = ?
Comp. = ?
Compare Register CC60R
Compare Register CC61R
Compare Register CC62R
T12_ST
Compare Shadow Register CC60SR
Compare Shadow Register CC61SR
Compare Shadow Register CC62SR

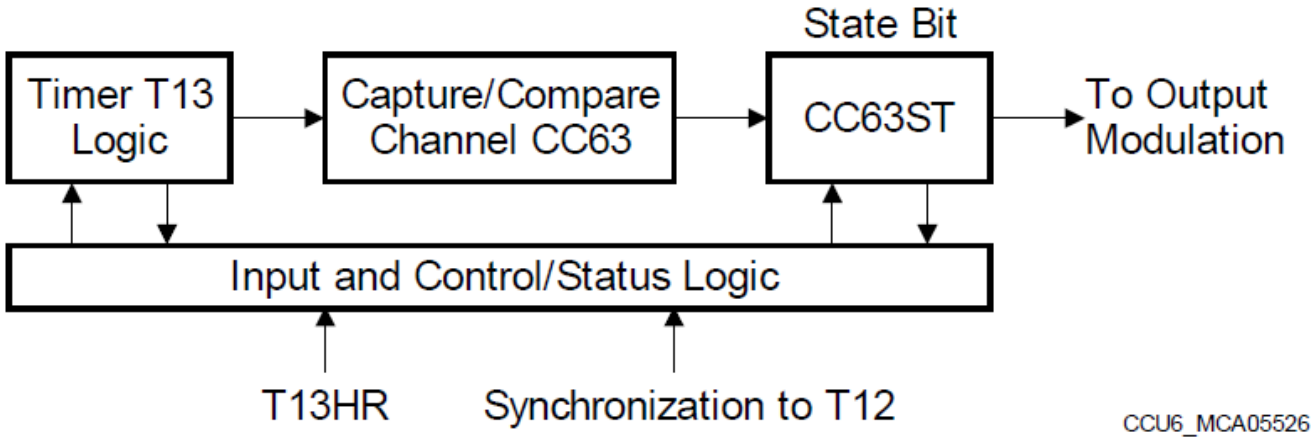
CCU6 (Capture and Compare Unit)



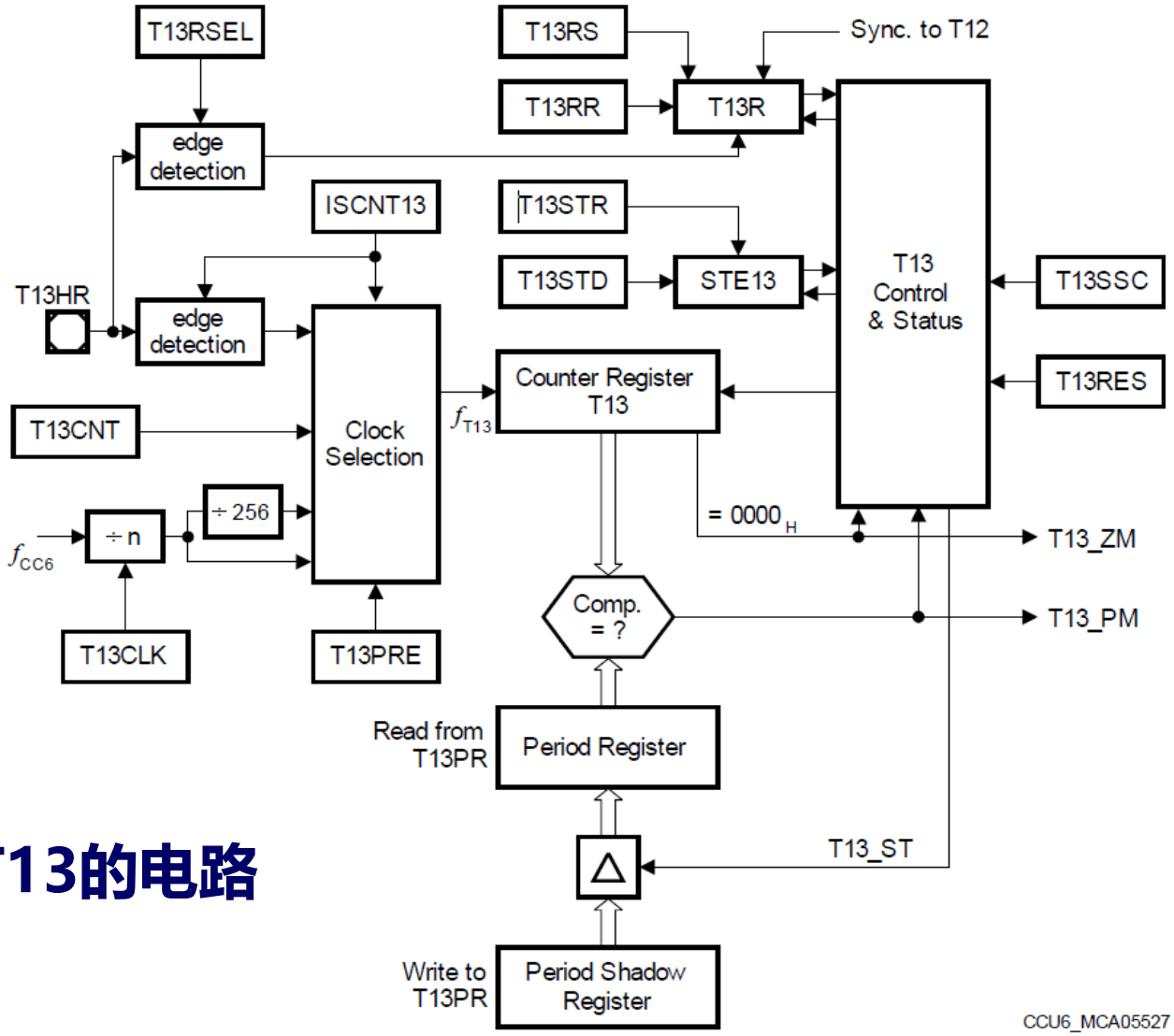
CCU6 (Capture and Compare Unit)



定时器T13模块只有输出比较功能，但是可以和T12联动

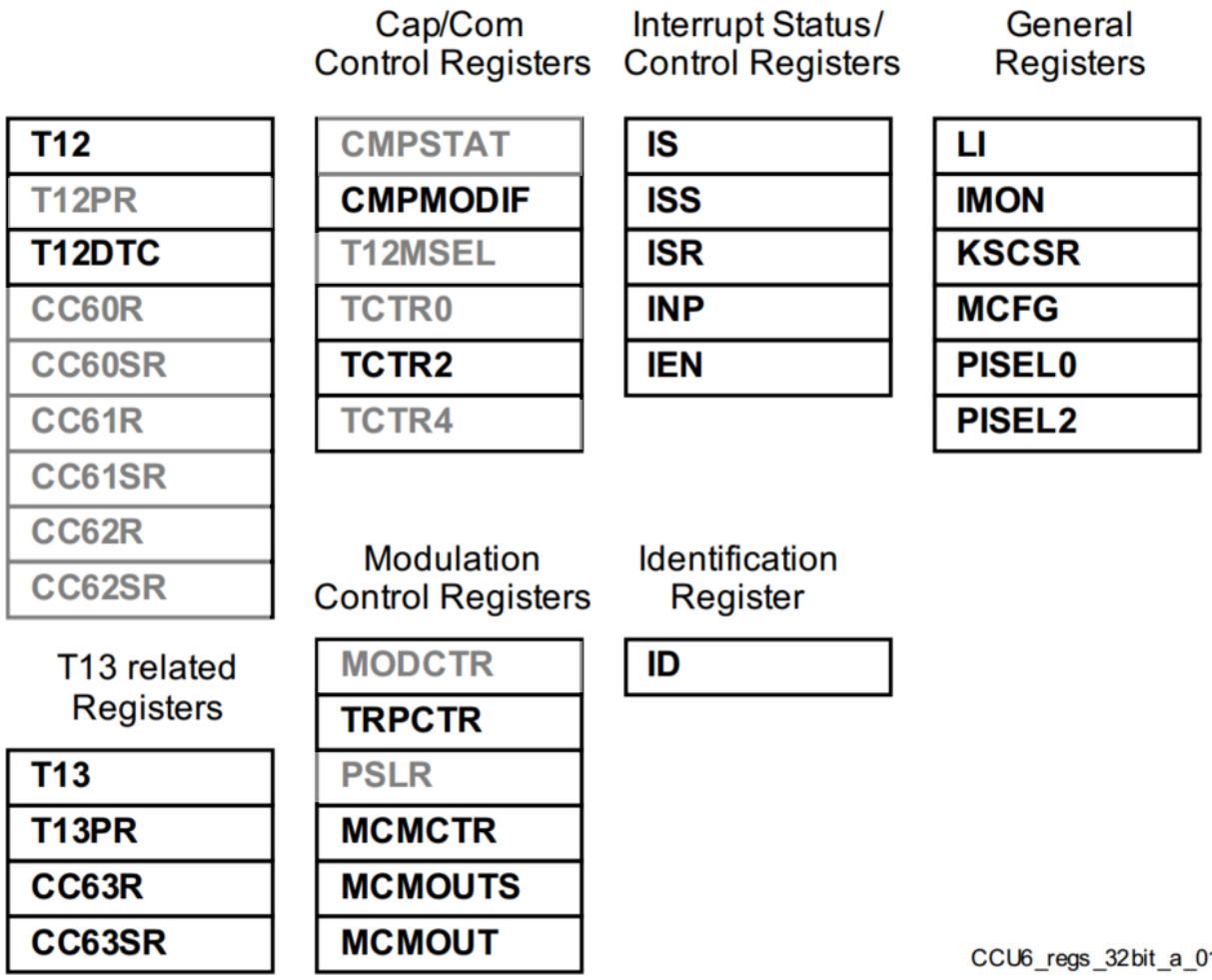


CCU6 (Capture and Compare Unit)



定时器T13的电路

可能用到的寄存器


Cap/Com Control Registers
Interrupt Status/ Control Registers
General Registers
T12
T12PR
T12DTC
CC60R
CC60SR
CC61R
CC61SR
CC62R
CC62SR
CMPSTAT
CMPMODIF
T12MSEL
TCTR0
TCTR2
TCTR4
IS
ISS
ISR
INP
IEN
LI
IMON
KSCSR
MCFG
PISEL0
PISEL2
Modulation Control Registers
Identification Register
T13 related Registers
MODCTR
TRPCTR
PSLR
MCMCTR
MCMOUTS
MCMOUT
ID
T13
T13PR
CC63R
CC63SR
CCU6_regs_32bit_a_01

PWM初始化函数（仅执行一次）

CCU6模块使能（CLC.B.DISR、CLC.B.DISS）

选择计数模式（TCTR0.B.CTM）

配置时钟频率（TCTR0.B.T12PRE、TCTR0.B.T12CLK）

设置工作在输出比较模式（T12MSEL.B.MSEL60）

设置输出比较信号隐性状态对应的电平状态（PSLR.B.PSL、CMPSTAT.B.COUT60PS）

设置频率和占空比（T12PR.B.T12PV、CC60SR.B.CCS）

使能影子传输（TCTR4.B.T12STR）

启动时钟（TCTR4.B.T12RS）

通道调制输出使能（MODCTR.B.T12MODEN）

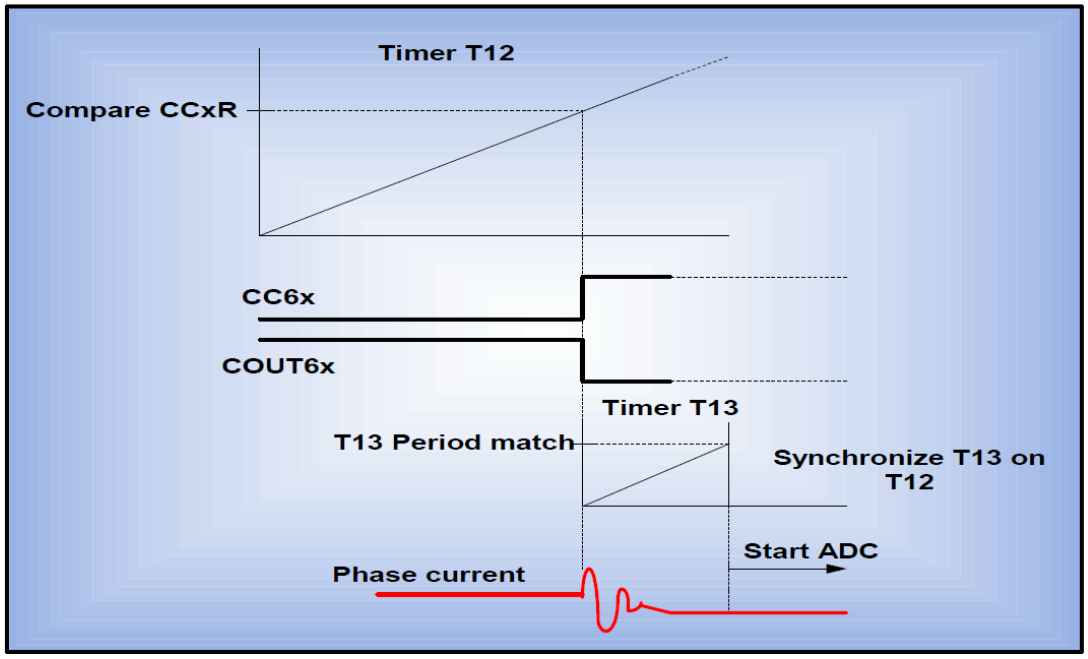
IO口设置为输出（IOCR）

（死区设置（T12DTC.B.DTM、T12DTC.B.DTE0））

CCU6（Capture and Compare Unit)

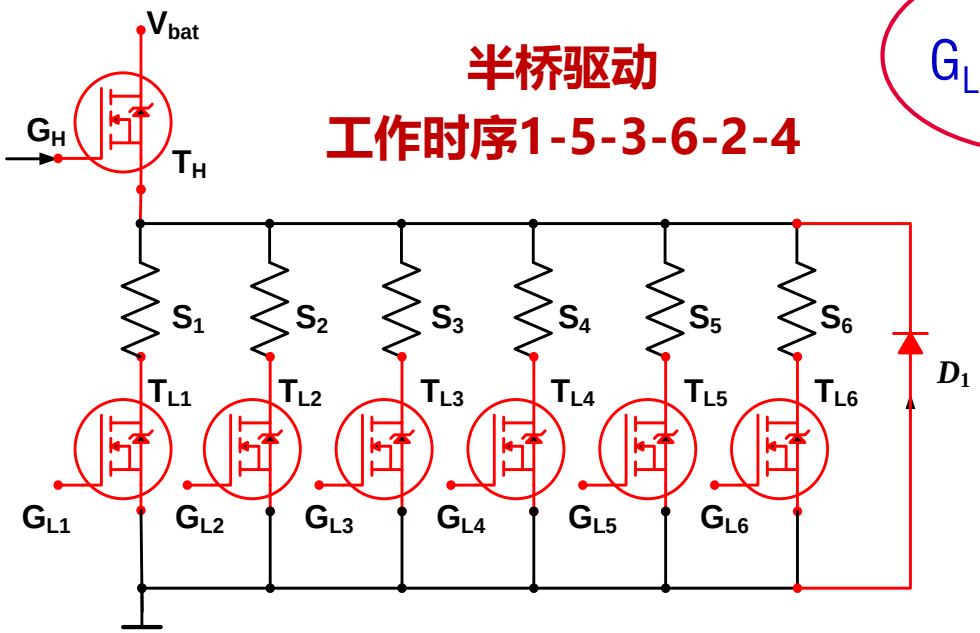
CCU6可以和ADC配合工作，由CCU6的输出比较作为外部事件触发AD转换，对执行器的电流采样通常采用这种模式。

- Synchronize T13 on T12
- T13 performs delay for stable measurement
- T13 period match triggers ADC



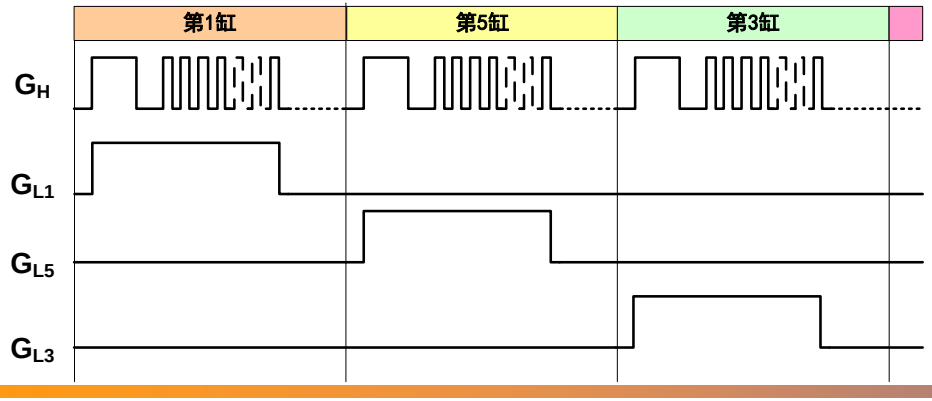
CCU6（Capture and Compare Unit)

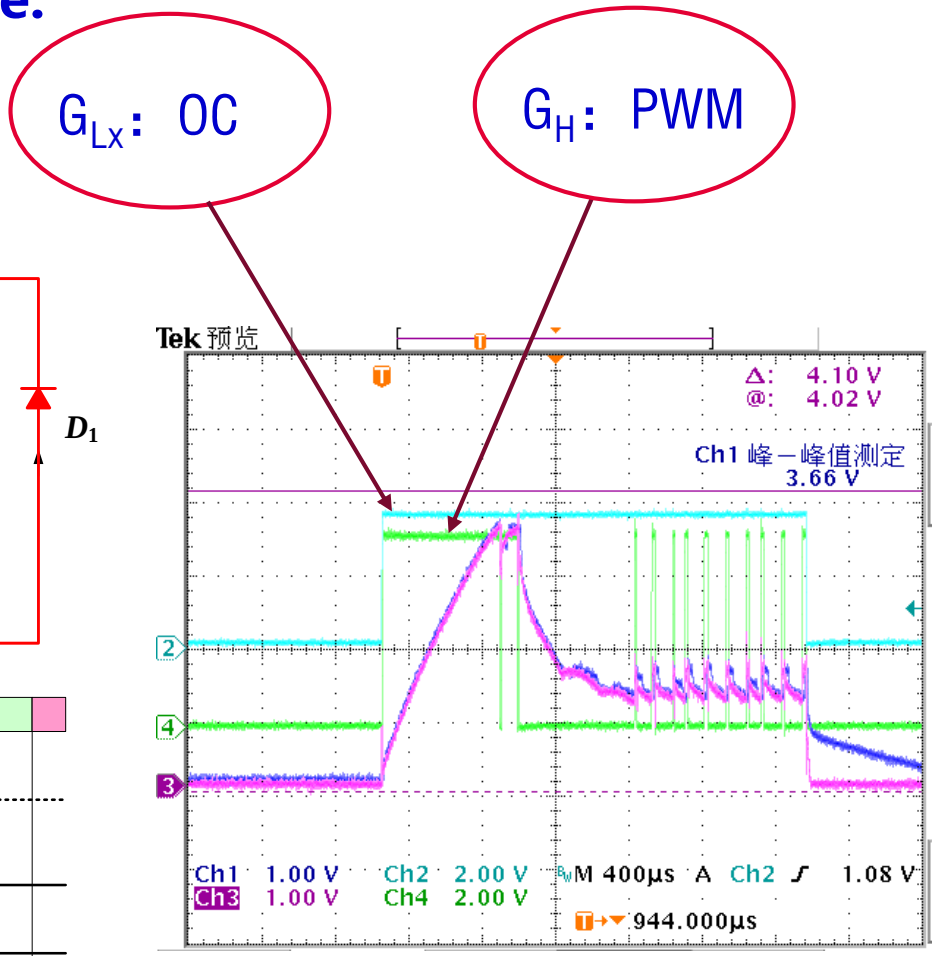
■电控燃油喷射的半桥驱动电路/ Half bridge drive is a combination of high side drive and low side drive.



半桥驱动

工作时序1-5-3-6-2-4

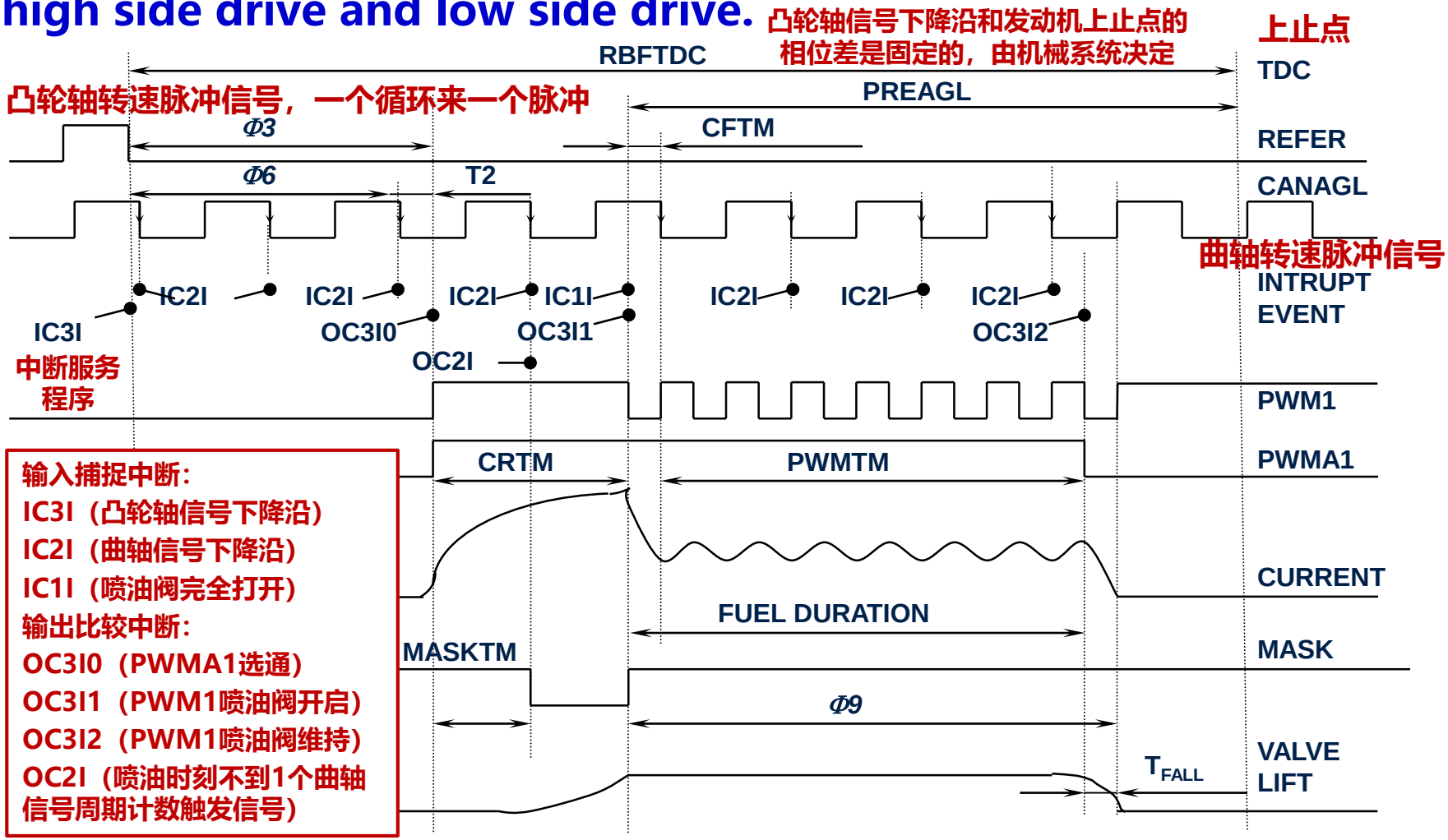




一次喷射过程

问：上述六缸发动机一个工作循环喷射几次?

■电控燃油喷射的半桥驱动电路/ Half bridge drive is a combination of high side drive and low side drive.

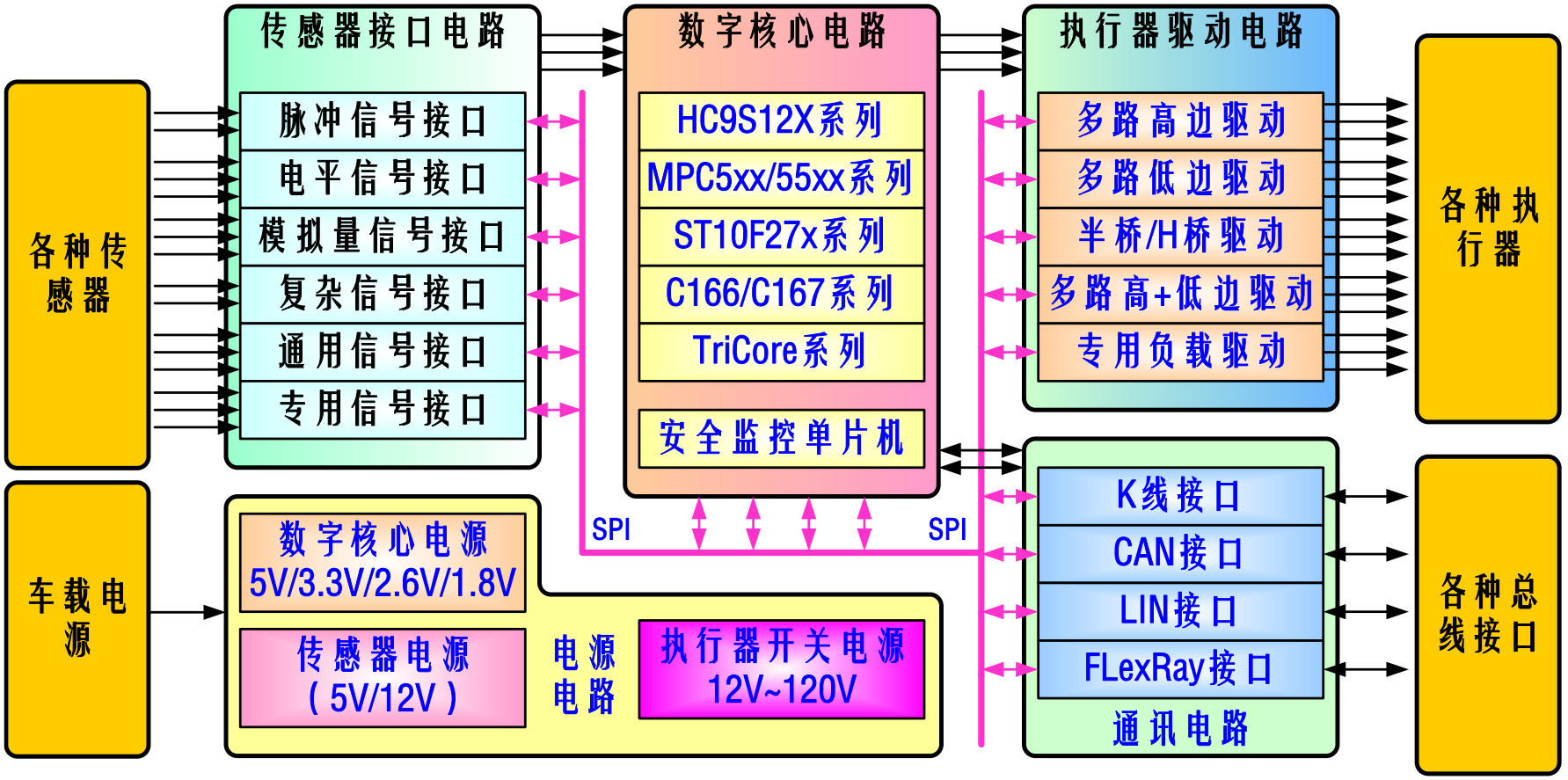


输入捕捉中断：
IC3I（凸轮轴信号下降沿）
IC2I（曲轴信号下降沿）
IC1I（喷油阀完全打开）
输出比较中断：
OC3I0（PWMA1选通）
OC3I1（PWM1喷油阀开启）
OC3I2（PWM1喷油阀维持）
OC2I（喷油时刻不到1个曲轴信号周期计数触发信号）

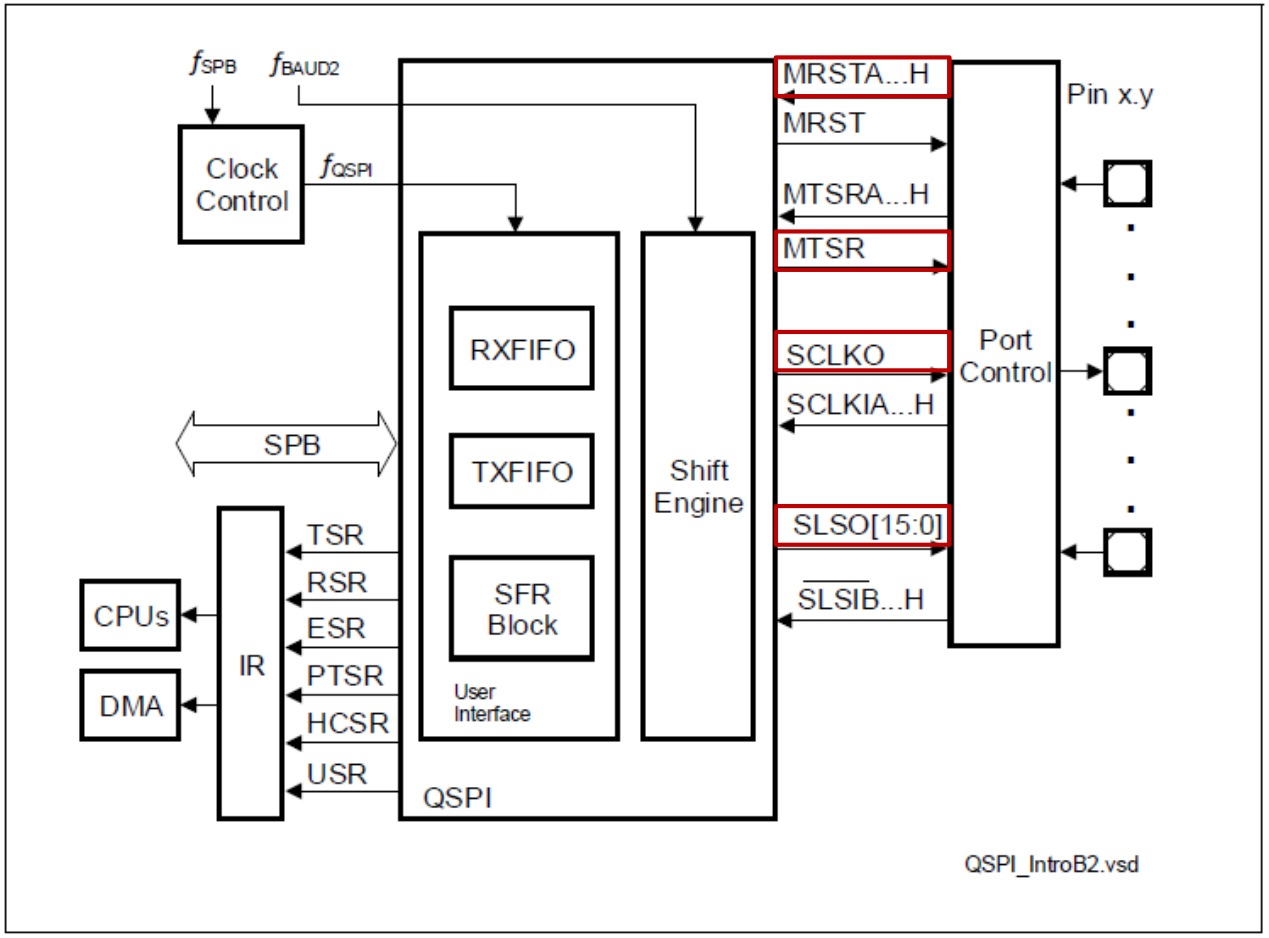
喷射控制的时序/ Injection sequence

- 4.2 CCU6
- 4.3 QSPI

QSPI通讯模块


传感器接口电路
脉冲信号接口
电平信号接口
模拟量信号接口
复杂信号接口
通用信号接口
专用信号接口
数字核心电路
HC9S12X系列
MPC5xx/55xx系列
ST10F27x系列
C166/C167系列
TriCore系列
安全监控单片机
执行器驱动电路
多路高边驱动
多路低边驱动
半桥/H桥驱动
多路高+低边驱动
专用负载驱动
各种传感器
各种执行器
车载电源
数字核心电源
5V/3.3V/2.6V/1.8V
传感器电源
(5V/12V)
电源电路
执行器开关电源
12V~120V
SPI
SPI
K线接口
CAN接口
LIN接口
FLexRay接口
通讯电路
各种总线接口

QSPI通讯模块



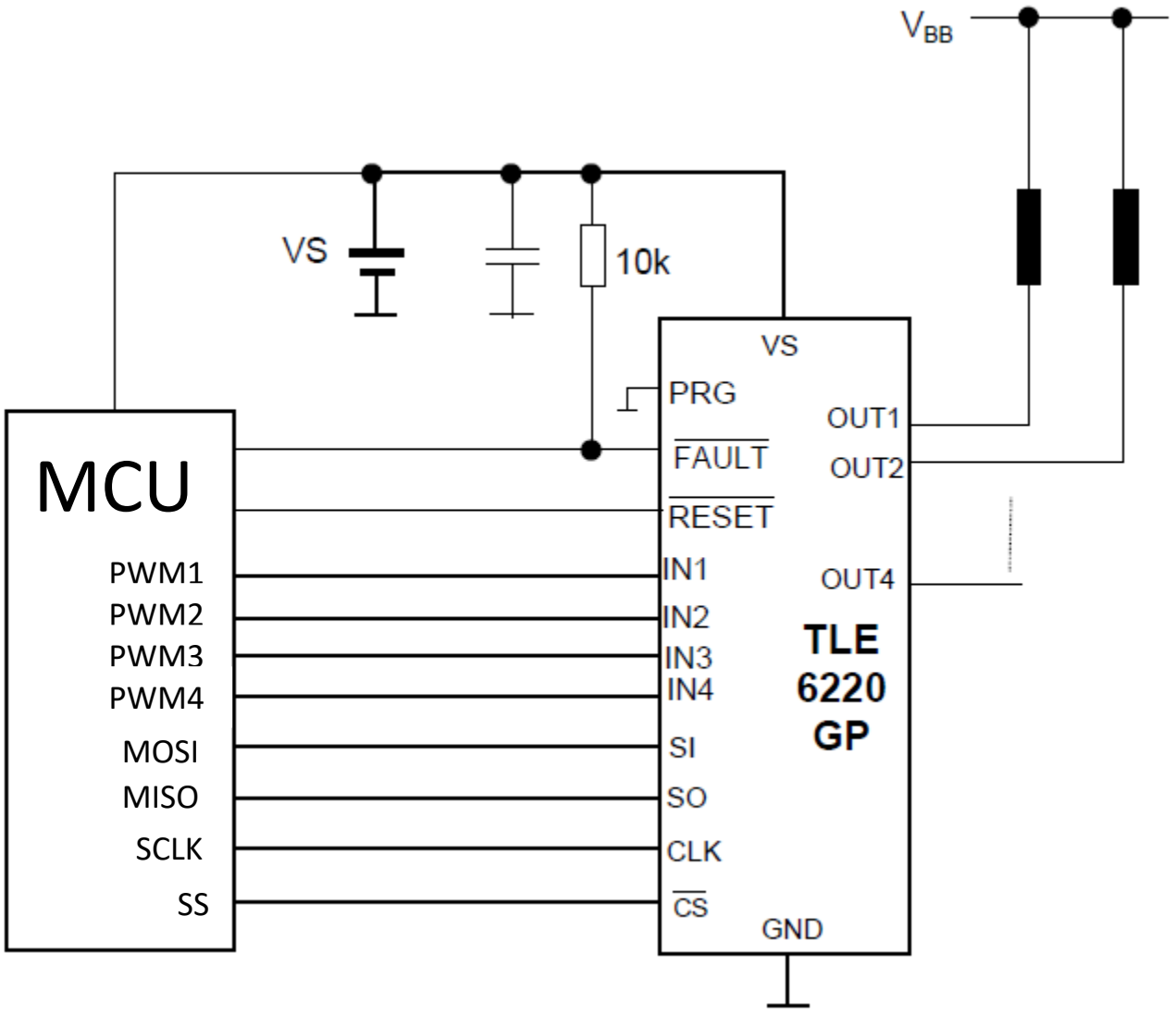
发送

接收

时钟

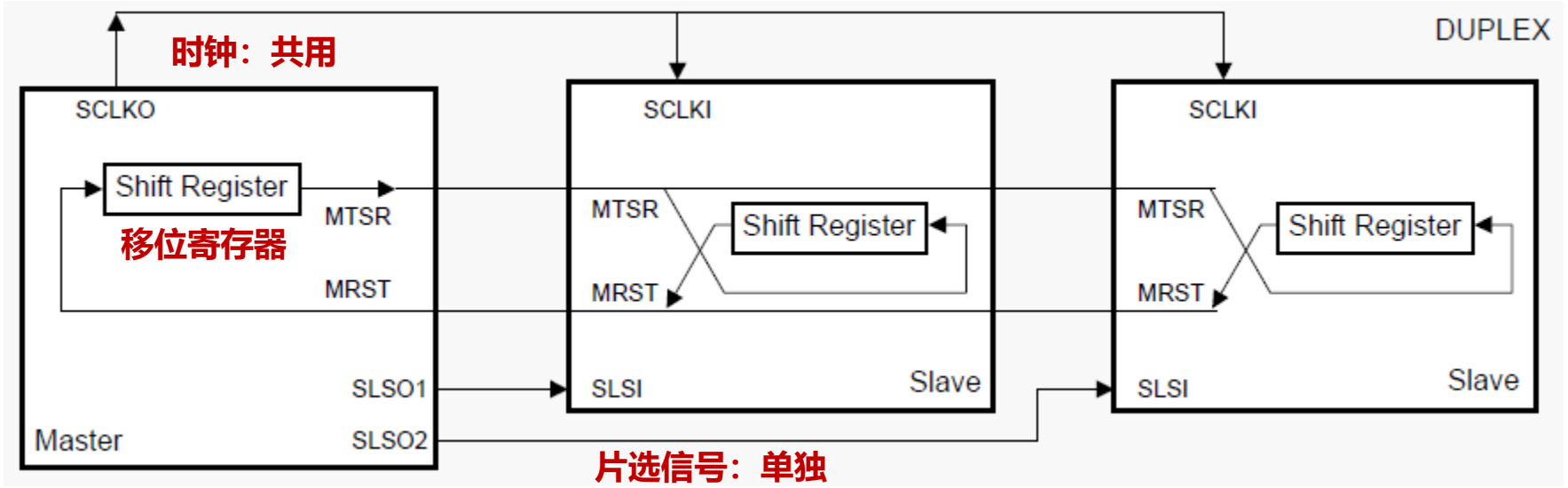
片选

QSPI通讯模块


VBB
VS
10k
VS
PRG
FAULT
RESET
IN1
IN2
IN3
IN4
SI
SO
CLK
CS
GND
OUT1
OUT2
OUT4
TLE
6220
GP
MCU
PWM1
PWM2
PWM3
PWM4
MOSI
MISO
SCLK
SS

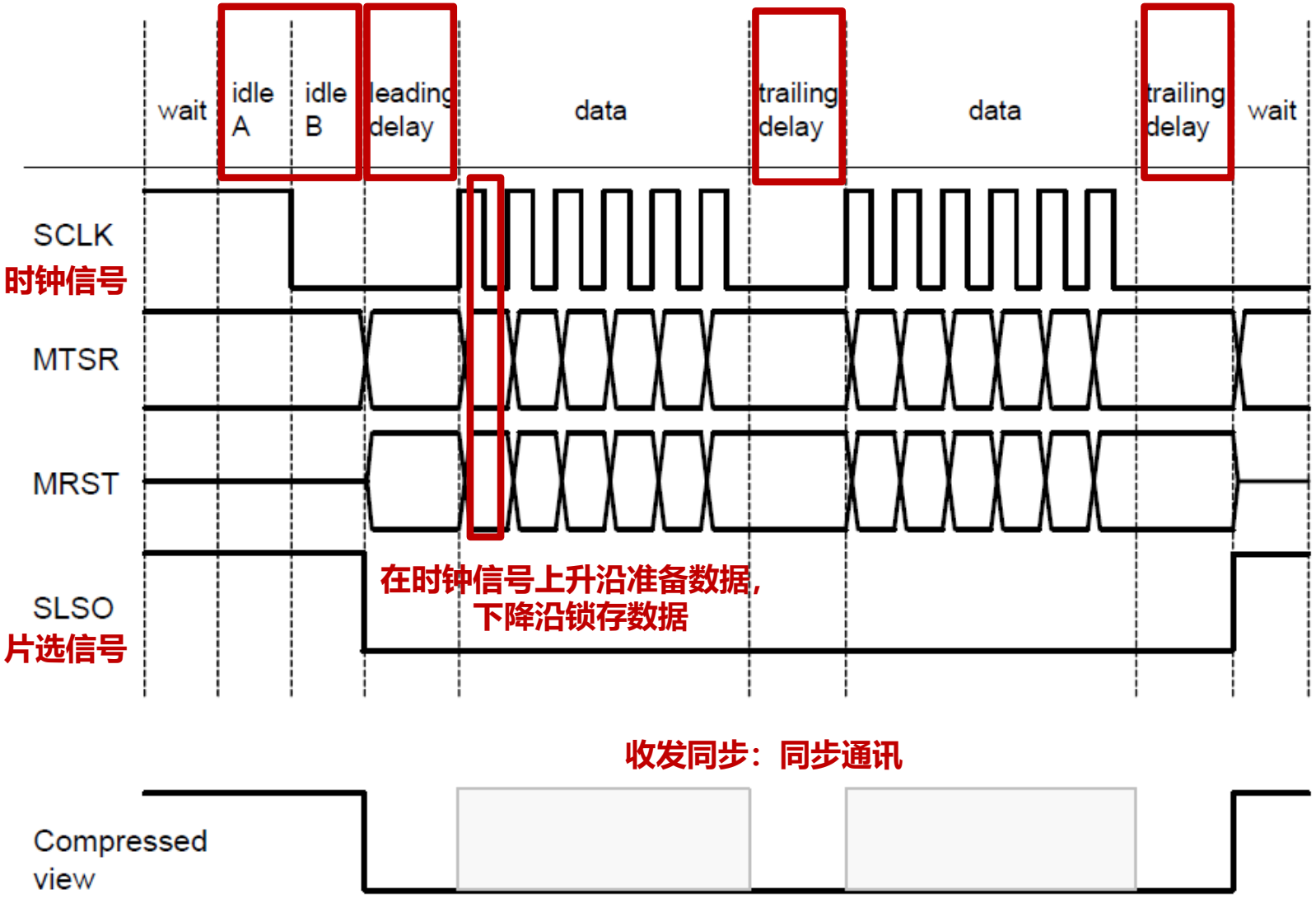
QSPI通讯模块

- New queued SPI (Synchronous Peripheral Interface) peripheral for AURIX
- QSPI concepts
- Transmission modes

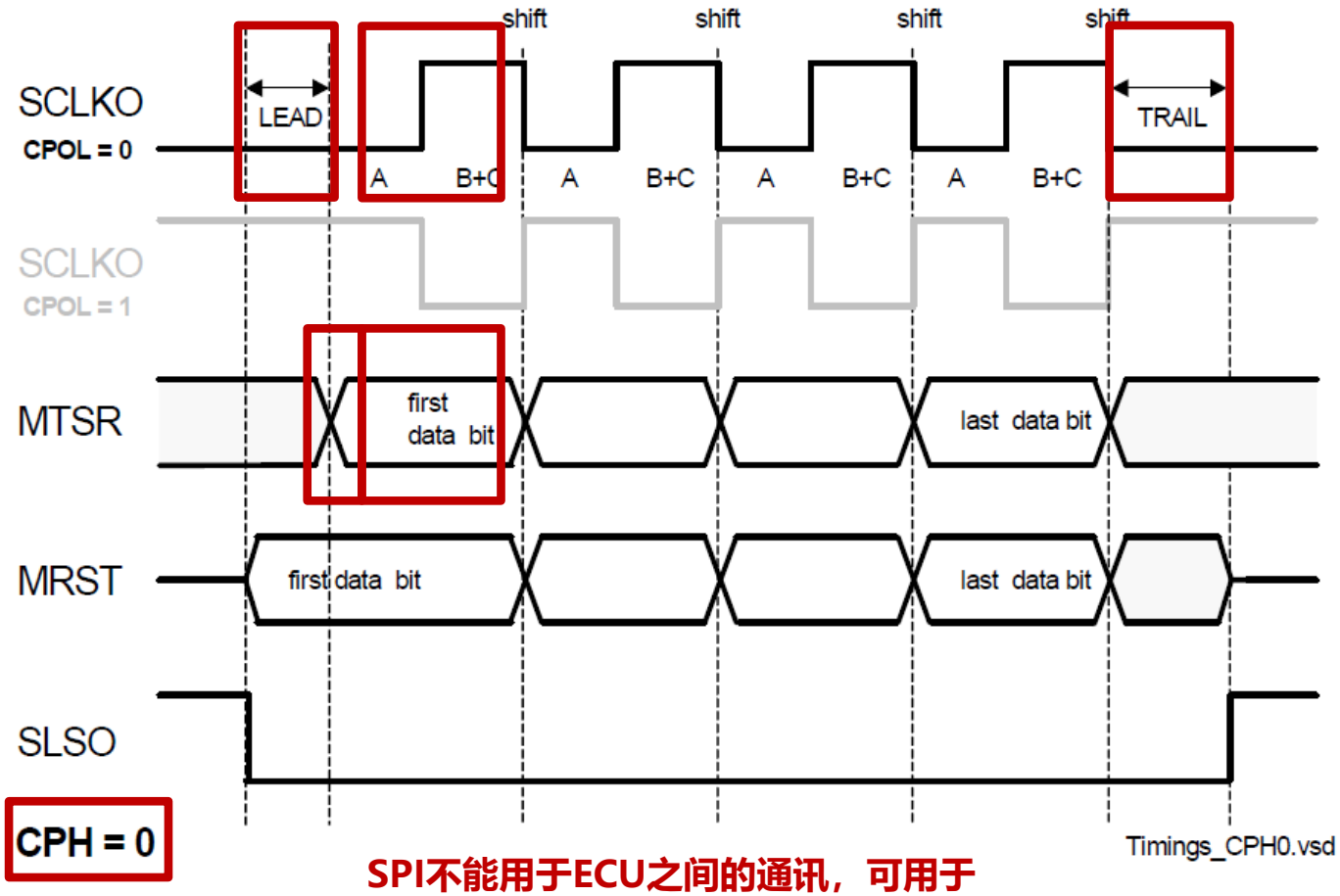


MTSR: Master Transmit Slave Receive
MRST: Master Receive Slave Transmit
SLSO: Slave Select Output

QSPI通讯模块


wait
idle A
idle B
leading delay
data
trailing delay
data
trailing delay
wait
SCLK
时钟信号
MTSR
MRST
SLSO
片选信号
在时钟信号上升沿准备数据，
下降沿锁存数据
收发同步：同步通讯
Compressed view

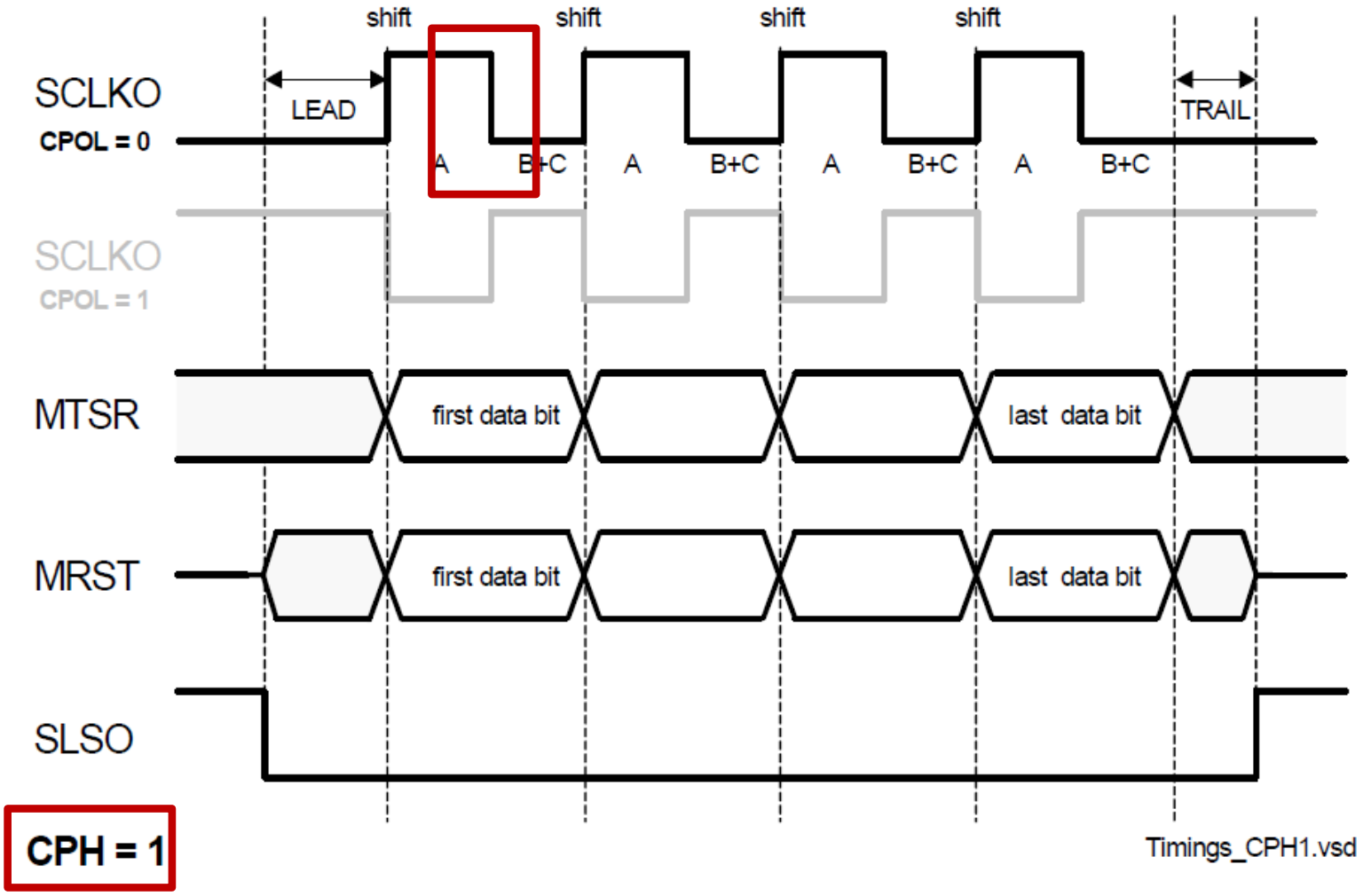
QSPI通讯模块



SPI不能用于ECU之间的通讯，可用于单片机和外围芯片的通讯

CPH	12	rw	**Clock Phase** Delay of one half SCLK clock cycle. 0_B Disabled 1_B Enabled

QSPI通讯模块


shift
shift
shift
shift
SCLKO
CPOL = 0
LEAD
TRAIL
A
B+C
A
B+C
A
B+C
A
B+C
SCLKO
CPOL = 1
MTSR
first data bit
last data bit
MRST
first data bit
last data bit
SLSO
CPH = 1
Timings_CPH1.vsd

QSPI通讯模块

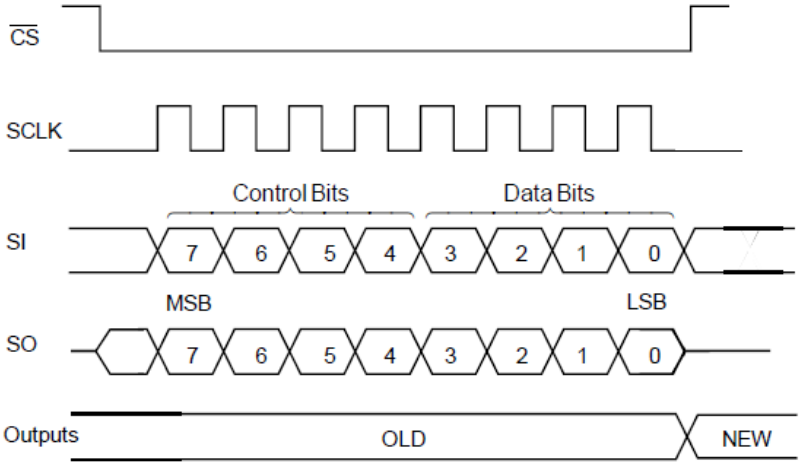
- TLE6220的SPI通讯协议
  - 高4位控制字、低4位数据字
  - 能够识别5种控制字



H：1
L：0

No.	Serial Input Byte	Function
1	LLLL XXXX	Only 'Full Diagnosis' performed. No change to output states.
2	HHLL XXXX	State of four parallel inputs and '1-bit Diagnosis' outputted.
3	HLHL XXXX	Echo-function of SPI; SI direct connected to SO
4	LLHH DDDD	IN1...4 and serial data bits 'OR'ed. 'Full Diagnosis' performed.
5	HHHH DDDD	IN1...4 and serial data bits 'AND'ed. 'Full Diagnosis' performed.

LLLL XXXX：正常诊断，不改变输出状态
HHLL XXXX：读取输入状态和1位诊断码
HLHL XXXX：SPI反射，接收=发送
LLHH DDDD：按位“或”操作，返回诊断码
HHHH DDDD：按位“与”操作，返回诊断码



Outline

- New queued SPI peripheral for AURIX
- QSPI concepts
- Transmission modes

New Queued SPI Peripheral in AURIX

- New features of queued SPI (QSPI) peripheral
  - Queued mode support
    - Handling of both **control** information and the **transmit data**
    - Enabling efficient data movement by DMA engines
    - Small FIFOs within peripheral to buffer data movement
  - Automatic hardware control of signal timing
    - Slave select signal activation
    - Slave select leading, trailing, and idle delay times
  - Support for high speed SPI communications
    - Up to **50 Mbit/s**
    - Programmable time quanta per bit (similar to CAN)
    - Overrun protection when receive FIFO is full

Additional SPI Features

- Some additional, unique features
  - Data lengths can be specified in **bytes** (instead of **bits**) for long data streams
  - Up to **16** slave select signals per SPI channel
  - Support for hardware parity checking
- Traditional SPI features are also supported
  - Programmable data lengths of **2 to 32 bits** (or **bytes**)
  - Transmit direction - most or least significant bit first
  - Clock polarity – clock idle high or low
  - Clock phase – data shift on leading or trailing clock edge
  - Interrupt generation on FIFO events and phase transitions

What is the “queue” in QSPI?

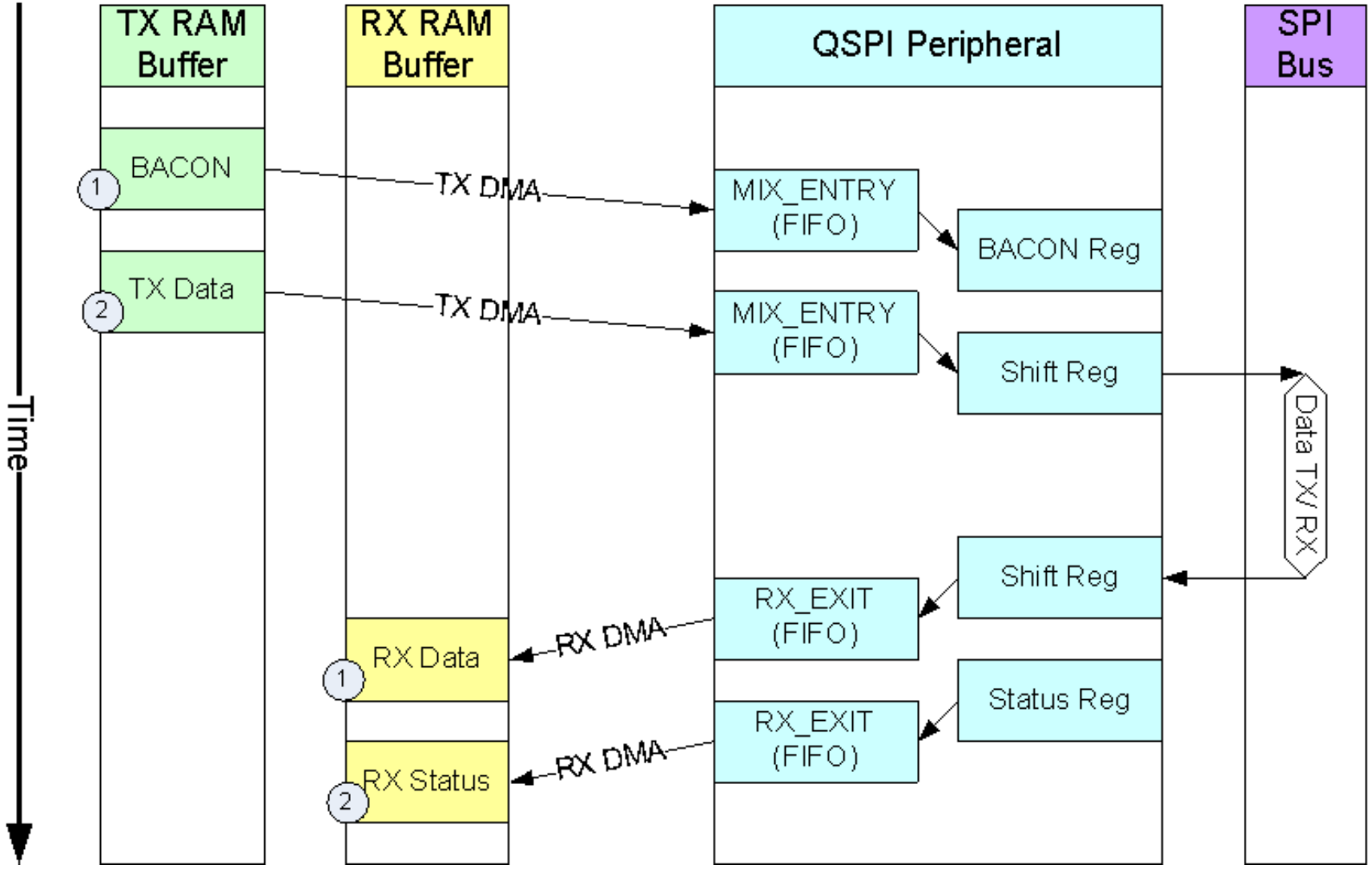
- The “queue” in QSPI is a buffer in general purpose RAM
  - The queue contains two types of data
    1. Configuration data (***how*** the data should be transmitted)
    2. Transmit data (***what*** data should be transmitted)
  - The simplest queue format interleaves the configuration and transmit data every other **32-bit word**
- Example of interleaved queue:

Byte Offset	Queue Content
0	CONFIG
4	TX DATA
8	CONFIG
12	TX DATA
16	CONFIG
20	TX DATA

Why Use a Queue?

- A queue enables efficient streaming of SPI data
  - The queue data can be easily transferred to the QSPI peripheral using a single DMA channel
  - The SPI data format and timing configuration can be changed automatically, on the fly, by the transfer of queue data (no need for CPU intervention)
  - A special FIFO mechanism has been implemented to direct incoming queue data to the appropriate destination (either a configuration register, or transmit shift register)

Sequence Diagram for Queue Data Transfers



Outline

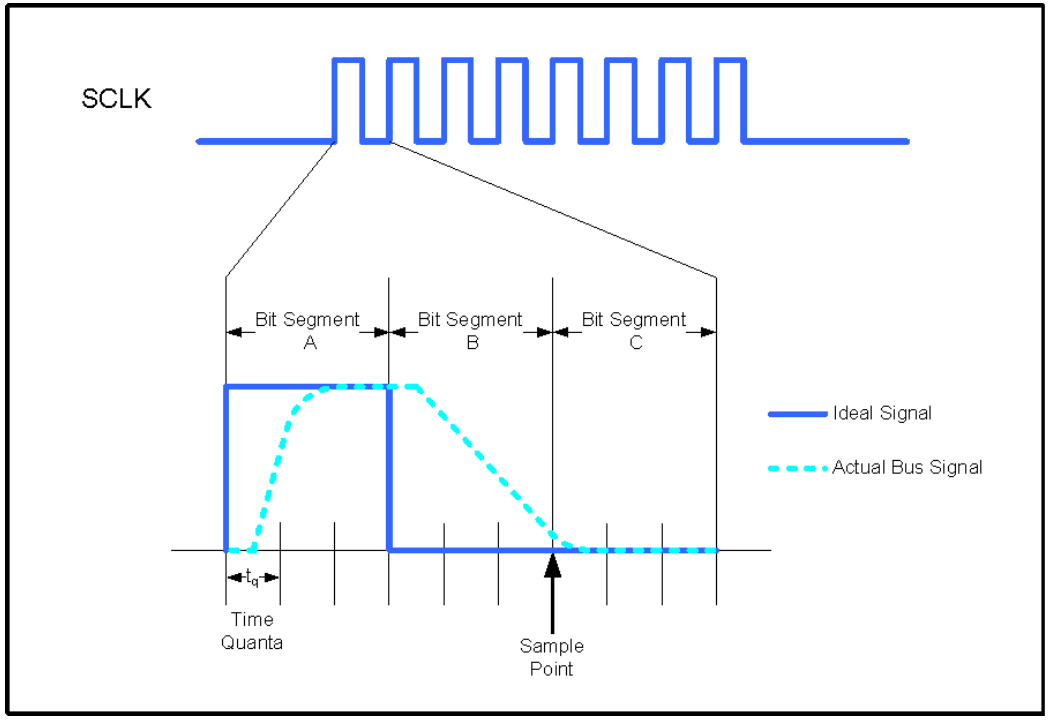
- New queued SPI peripheral for AURIX
- QSPI concepts
- Transmission modes

QSPI Time Quanta

- High SPI clock rates (for example, more than **50MHz**) introduce technical challenges:
  - The signal rise and fall times become large relative to the SPI clock period
  - SPI clock rise and fall times may not always be symmetrical
- QSPI provides a mechanism to divide the SPI clock signal into smaller time quanta (similar to CAN)
  - Allows precise control of the clock high and low times, and sample point

QSPI Time Quanta (2)

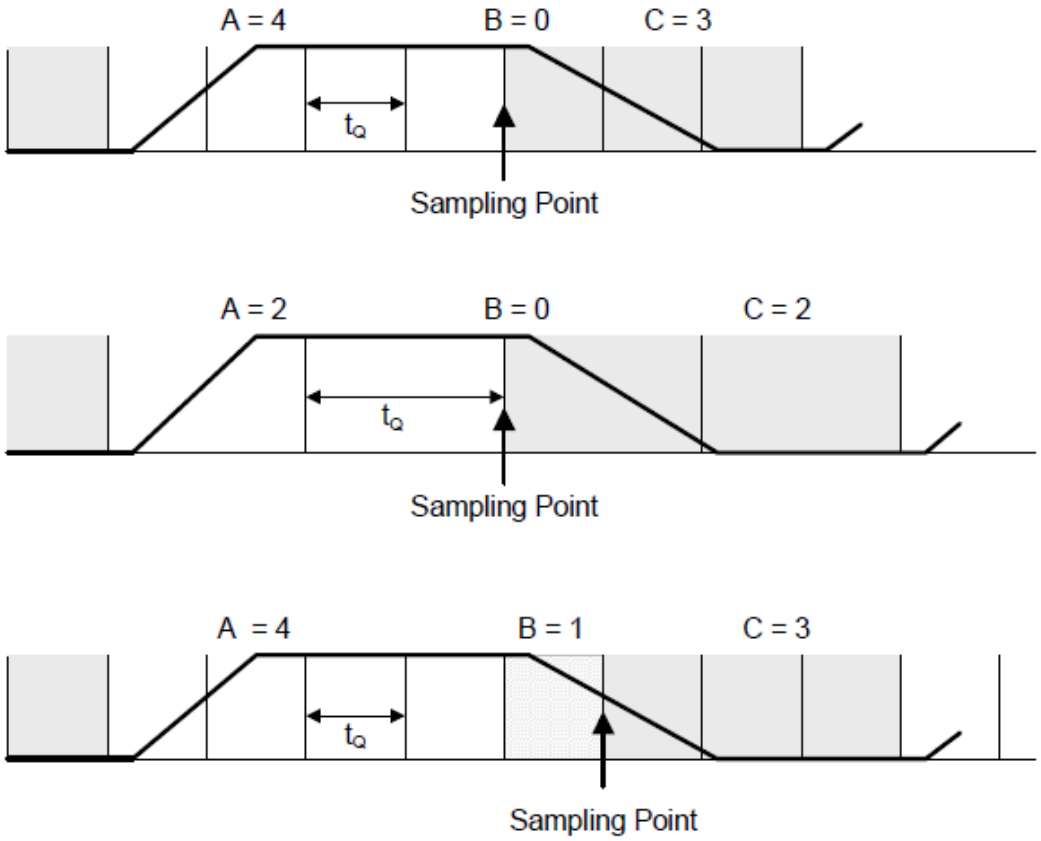
- For more precise signal timing, the SPI signals are divided into smaller **time quanta** and **bit segments**



- For simple SPI use cases, just two time quanta are used for each SPI clock period (A=1, B=0, C=1) → equal high and low times, sample point in middle of clock period

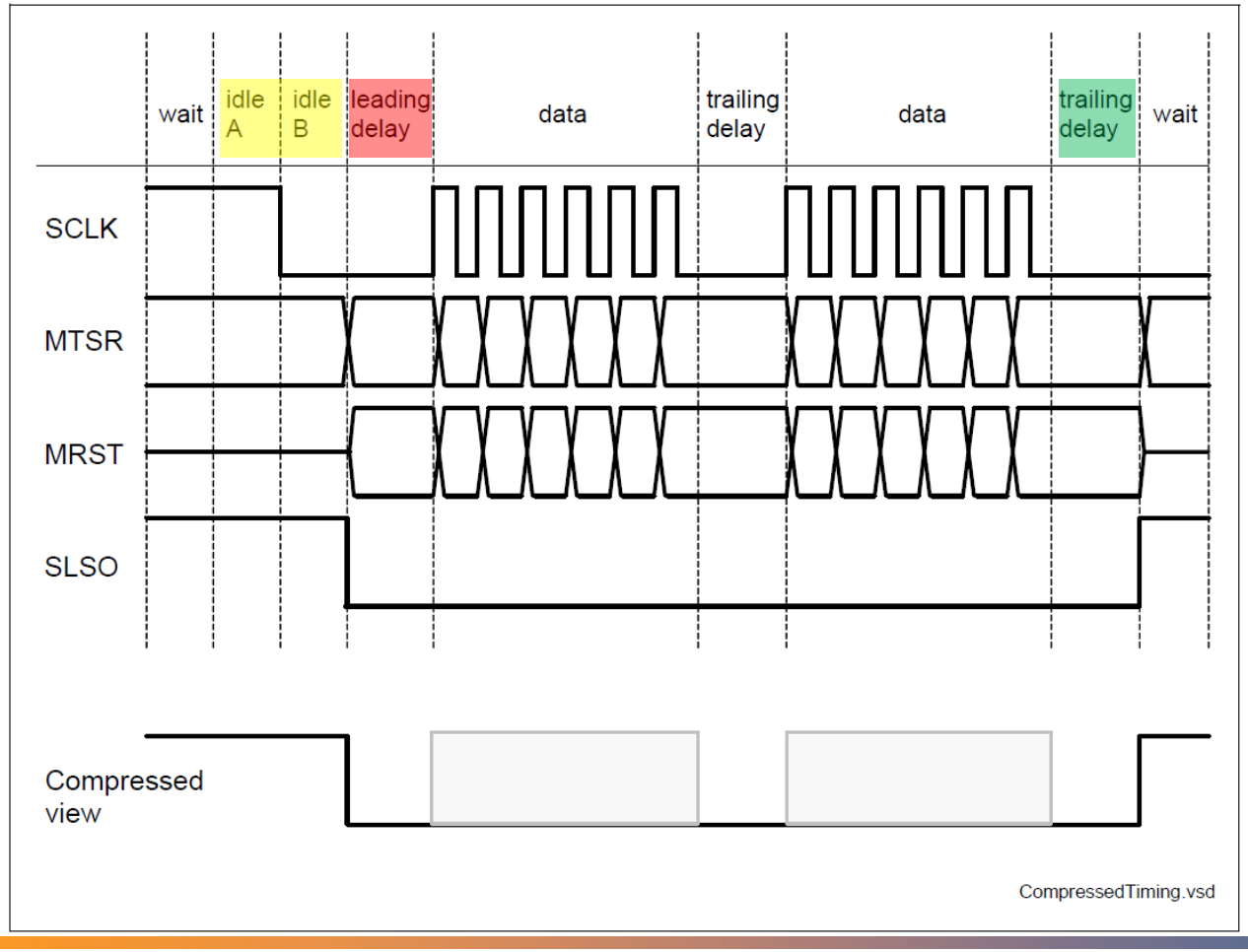
QSPI Time Quanta (2)

- **时钟配置**
  - **时钟的占空比、接收信号的采样点的配置**
  - **ECON.A、B和C三个寄存器配置**



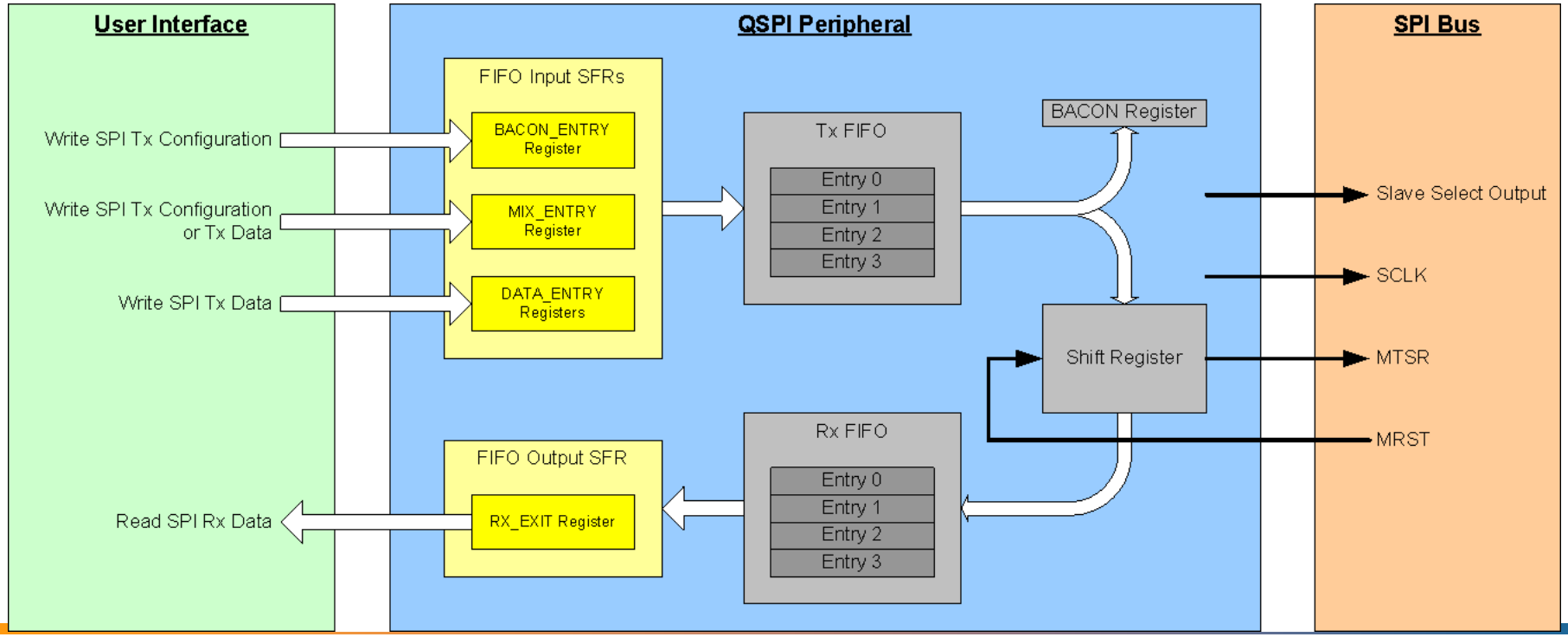
SPI Signal Delay Timing

- QSPI provides precise control over the idle, leading, and trailing delays

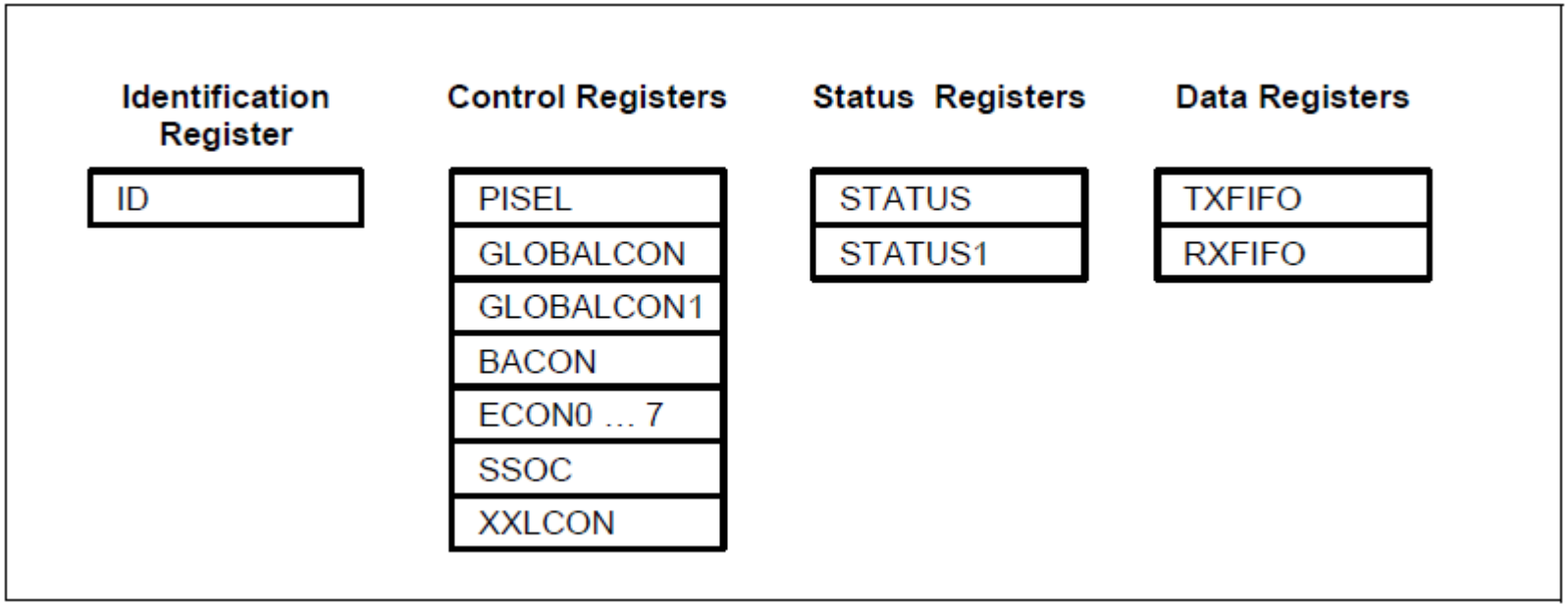


QSPI FIFO User Interfaces

- Writes to ***any*** "ENTRY" register delivers a new entry into TX FIFO
  - **BA**sic **CON**figuration (BACON) register can only be written through FIFO
- Read of RX FIFO only occurs through "EXIT" register



QSPI 寄存器


Identification Register
ID
Control Registers
PISEL
GLOBALCON
GLOBALCON1
BACON
ECON0 … 7
SSOC
XXLCON
Status Registers
STATUS
STATUS1
Data Registers
TXFIFO
RXFIFO

BACON and ECONz Registers

- The format of the SPI transmission is controlled by both the **BACON** and **ECONz** registers
  - BACON = **BA**sic **CON**figuration Register
  - ECONz = **E**xtended **CON**figuration Registers (z= 0..7)

- The channel select bit field of the BACON register selects which ECON register will be used

- Together, both registers define how data will be transmitted to each SPI slave device

BACON and ECONz Registers

BACON

Defines the basic configuration parameters for the current slave select. It can be read by software, or written through a write to the TXFIFO.

(>> **Table 20-2** register overview)

BACON

Basic Configuration Register **(18_H)** **Reset Value: $0F87\ 1C71_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CS				DL					BYTE	MSB	UINT	PARTYP	TRAIL		
rh				rh					rh	rh	rh	rh	rh		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TPRE			LEAD			LPRE			IDLE			IPRE			LAST
rh			rh			rh			rh			rh			rh

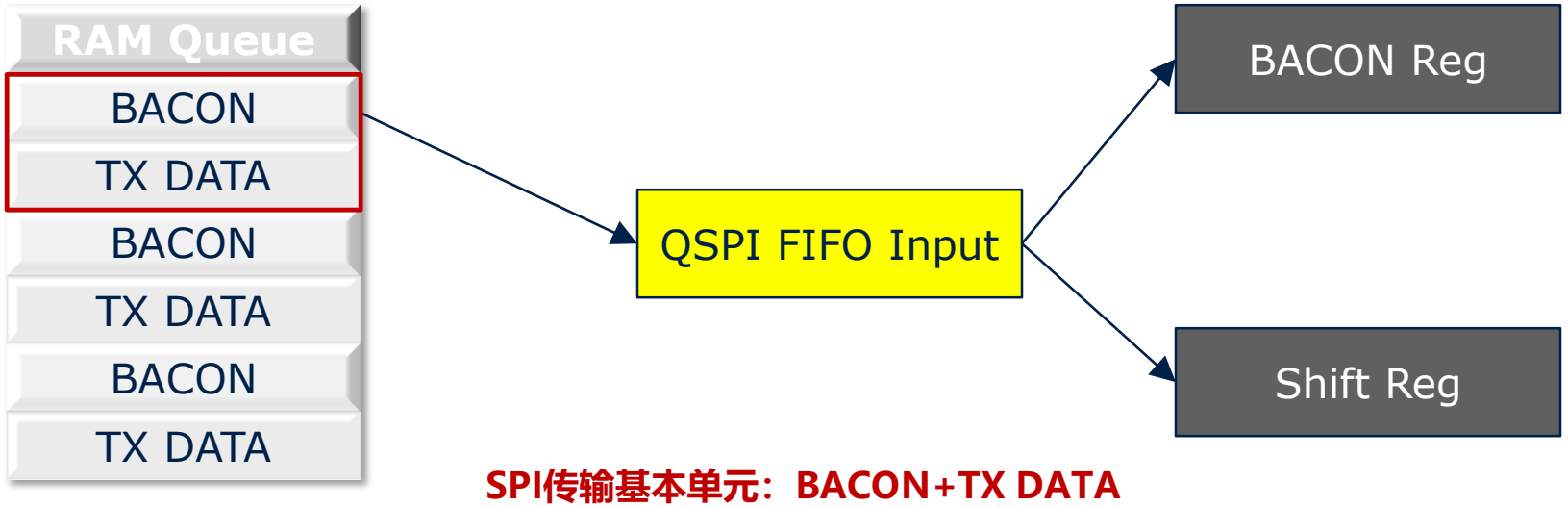
BACON Register Fields

- BACON register: ***Dynamic*** SPI parameters

Field	Range	Description
Last	0 or 1	•0 = next data word will not end frame •1 = next data word is last of frame, slave select will be disabled 决定了片选信号 SLSO
IDLE, LEAD, TRAIL delays	Formula	•Defines idle, leading, and trailing delay times
Parity type	0 or 1	•0 = even parity •1 = odd parity
User interrupt	0 or 1	•1 = generate user interrupt at PT1 (phase transition) event
MSB	0 or 1	•0 = least significant bit transmitted first •1 = most significant bit transmitted first
Byte	0 or 1	•0 = data length is in bits •1 = data length is in bytes
Data length	2-32	•SPI word size in bits or bytes
Channel select	0-15	•Activates corresponding slave select output •Also selects corresponding extended configuration register ECONz (note that CS values 8..15 wrap around and overlap ECON 0..7)

BACON Register

- The BACON register value can ***only*** be programmed through the transmit FIFO
- BACON values and transmit data are ***streamed*** from the RAM queue to the QSPI FIFO
  - BACON values ***dynamically*** change the way in which SPI data is transmitted



SPI传输基本单元：BACON+TX DATA

BACON and ECONz Registers

ECON

Configuration extensions for channels 0 to 15. Register x defines several timing characteristics for two channels: z and z+8.

(>> **Table 20-2** register overview)

(>> **Figure 20-12**, **Figure 20-13** timing diagrams)

ECONz (z = 0 - 7)

Configuration Extension z **($20_H + z^*4_H$)** **Reset Value: 0000 1450$_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BE		0													
rw	rw	r													

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	PAR EN	C POL	CPH	C		B		A		Q					
r	rw	rw	rw	rw		rw		rw		rw					

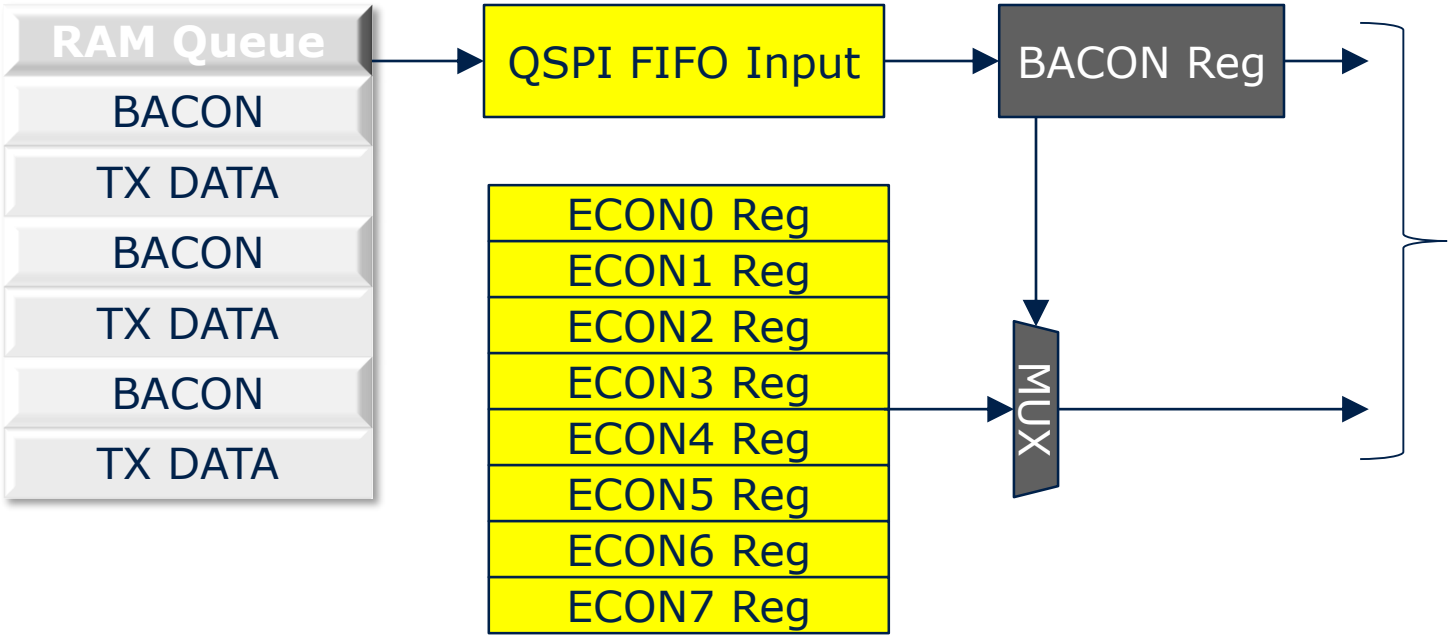
ECONz Register Fields

- ECON 0..7 registers:

Field	Range	Description
Q	1..64	•Time quanta divider for the channel
A	1..4	•Bit segment A width in time quanta units
B	0..3	•Bit segment B width in time quanta units
C	0..3	•Bit segment C width in time quanta units •If B=0, C is set to 1 by hardware
Clock phase	0 or 1	•0 = no delay in SPI clock •1 = delay SPI clock by one half cycle
Clock polarity	0 or 1	•0 = SPI clock idle low level •1 = SPI clock idle high level
Parity enable	0 or 1	•0 = parity disabled •1 = parity enabled

ECONz Registers

- The ECON 0..7 registers can be programmed directly
- The ECONz register values are typically ***static***, and programmed just once at initialization
- The active ECONz register is selected by the BACON channel select value



Outline

- New queued SPI peripheral for AURIX
- QSPI concepts
- Transmission modes

程序框架

SPI 通讯初始化函数：

1. 使能模块
2. 设置时钟、模式、缓冲区、输入输出通道、引脚、中断；
3. GLOBALCON_EN置1运行；
4. 相关引脚功能设置

主函数：

1.PWM初始化，输出PWM波
2.SPI初始化
3.SCI初始化
4.死循环中周期性调用SPI发送接收函数

SPI发送接收函数：

1. 将configuration 信息写到MIXENTRY
2. 将发送数据写到MIXENTRY
3. 等待发送完成（查询STATUS_TXFIFOLEVEL的数值）
4. 等待接收完成（查询STATUS_RXFIFOLEVEL的数值）
5. 从RXEXIT读取接收信息，储存在全局变量中方便查看
6. 将接收到的信息通过串口发送到电脑查看（利用第五次实验程序）

Transmission Modes

- QSPI offers a number of user selectable transmission modes
  - Short data mode
  - Long data mode
  - Continuous data mode
  - Single configuration – multiple frames

基本传输模式

组合传输模式

Short Data Mode

- Short data mode is characterized by frames with a single word transmission of 2 to 32 ***bits***
  - Mode is selected by BACON.BYTE = 0 (data length is in bits) and BACON.LAST = 1 (data is last in frame)
  - Complete transmission is defined by one 32-bit BACON configuration, and one 32-bit transmit data word

- Example queue layout for short data mode:

基本传输模式1

BACON.BYTE=0
BACON.LAST=1
一个BACON+一个DATA

Byte Offset	Queue Content
0	BACON (BYTE = 0)
4	TX DATA (2..32 bit)
8...	...more queue entries...



Long Data Mode

- Long data mode is characterized by frames with a transmission of 2 to 32 ***bytes***
  - Mode is selected by BACON.BYTE = 1 (data length is in bytes) and BACON.LAST = 1 (data is last in frame)
  - Queue transmit data sized according to number of bytes defined by data length (may need padding to 32-bit width)

- Example queue layout for data length = 7 bytes:

Byte Offset	Queue Content
0	BACON (BYTE = 1)
4	TX DATA BYTES 0..3
8	TX DATA BYTES 4..6 (padded)
12…	…more queue entries…



Compressed View of SPI Bus

基本传输模式2

BACON.BYTE=1
BACON.LAST=1
一个BACON+最多8个DATA

Continuous Data Mode

- Continuous data mode is characterized by frames with ***any arbitrary data length***
  - Frame started with BACON.LAST = 0 (**slave select remains active after data is transmitted**)
  - Frame ended by transmitting last data with BACON.LAST = 1 (data is last in frame, slave select goes inactive)
  - Continuous mode can be used with either short or long data
- Example queue layout for a 63-bit transmission:

Byte Offset	Queue Content
0	BACON (LAST=0, DL=32-bit)
4	TX DATA (32-bit)
8	BACON (LAST=1, DL=31-bit)
12	TX DATA (31-bit)
16	TX DATA BYTES 0..3



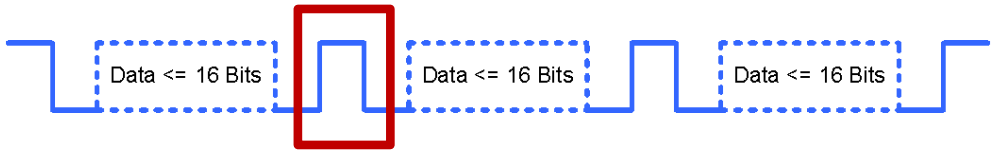
组合传输模式1

若干个BACON.LAST=0的基本单元 +BACON.LAST=1的基本单元

Single Configuration – Multiple Frames

- Single configuration – multiple frames allows an arbitrary number of frames to be sent to a slave using just one configuration
  - Provides a way to send many commands to a slave without the need for reconfiguration between each frame
  - Data length limited to a **maximum of 16-bit**
- Example queue layout for 3x 16-bit (or smaller) TX data:

Byte Offset	Queue Content
0	BACON (DL<=16-bit)
4	TX DATA (<=16-bit + padding)
8	TX DATA (<=16-bit + padding)
12	TX DATA (<=16-bit + padding)



组合传输模式2

BACON.BYTE=0或1
BACON.LAST=0
BACON.DL<=16bit
若干个连续的TX Data

Use Case Examples

- Example use case 1:
  - Multiple SPI slaves (assume 3)
  - Slaves require frames with 8, 16, or 32-bit data
- Possible QSPI configuration: Short data mode
  - Note: Assumes ECONz registers have been programmed already with correct time quanta for each slave baud rate

Byte Offset	Queue Content
0	BACON (CS=slave 0, LAST=1, DL=32-bit)
4	TX DATA (32-bit)
8	BACON (CS=slave 1, LAST=1, DL=8-bit)
12	TX DATA (8-bit + padding)
16	BACON (CS=slave 2, LAST=1, DL=16-bit)
20	TX DATA (16-bit + padding)



Use Case Examples

- Example use case 2:
  - Multiple slaves (assume 3)
  - Slaves require frames with 8, 16, or **33-bit** data
- Possible QSPI configuration: Short data mode + Continuous

Byte Offset	Queue Content
0	BACON (CS=slave 0, **LAST=0**, DL=16-bit)
4	TX DATA (16-bit + padding)
8	BACON (CS=slave 0, **LAST=1**, DL=17-bit)
12	TX DATA (17-bit + padding)
16	BACON (CS=slave 1, LAST=1, DL=8-bit)
20	TX DATA (8-bit + padding)
24	BACON (CS=slave 2, LAST=1, DL=16-bit)
28	TX DATA (16-bit + padding)

Q: Why not 32 + 1 bit here?

A: DL must be >= 2

DMA → MIX_ENTRY

Use Case Examples

- Example use case 3:
  - Slave requires a 96-bit frame

- Possible QSPI configuration: Long data mode

Byte Offset	Queue Content
0	BACON (CS=slave 0, LAST=1, DL=12 **bytes,** BYTE=1)
4	TX DATA (Bytes 0..3)
8	TX DATA (Bytes 4..7)
12	TX DATA (Bytes 8..11)



Q: Could short data mode be used here?

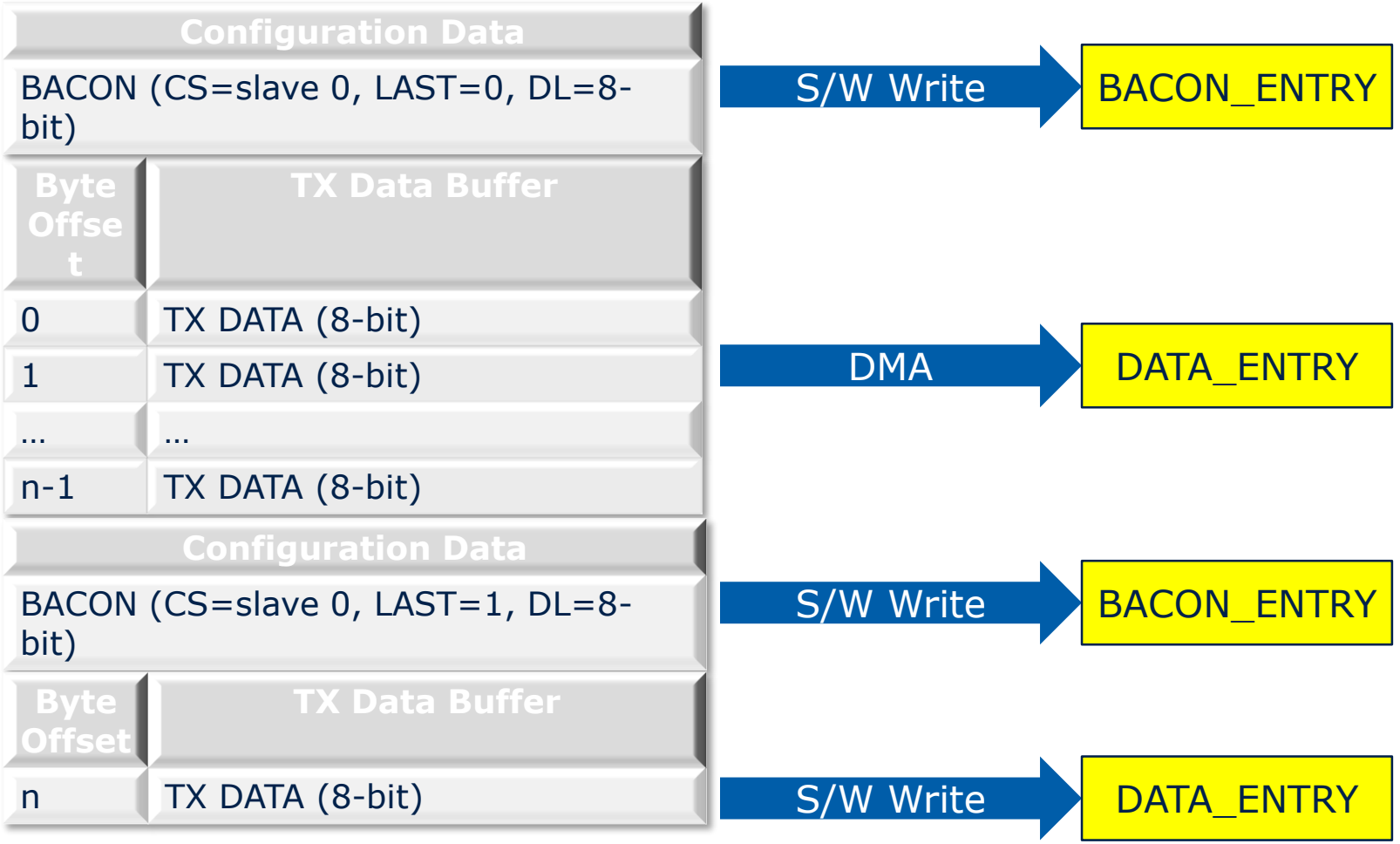
A: Yes, in combination with continous mode

Use Case Examples

- Example use case 4:
  - Full AutoSAR SPI driver
  - Configuration and transmit data managed from ***two separate RAM buffers***
    - With AutoSAR SPI concept, configuration and transmit data are ***not*** interleaved within the same buffer
    - The required SPI configurations are defined using **AutoSAR configuration tools**, held in **separate buffer from TX data**
  - Frames can be any arbitrary length
- Possible QSPI configuration: **Software managed configuration**
  - The AutoSAR SPI approach is very flexible, but restrictive when attempting to apply many of the hardware acceleration features of QSPI
  - A fully compliant AutoSAR SPI driver will likely fall back to a software driven implementation

Use Case Examples

- Example of software managed configuration with n x 8-bit TX data



小结

- 基本概念：PWM信号、输入捕捉Input Capture、输出比较Output Compare
- 边沿对齐Edge Aligned技术方式、中心对齐Center Aligned计数方式
- Zero Match, Period Match, Compare Match
- 四种输入捕捉模式Mode 1~4
- 输入捕捉：CC6xR、CC6xSR的作用是什么?
- 输出比较：CC6xR、CC6xSR、CC6xST的作用是什么？CC6xSR和CC6x、COUT6x有何关联?
- 边沿对齐输出比较模式的工作原理
- 中心对齐输出比较模式的工作原理
- 死区补偿模式（边沿对齐 或 中心对齐）的工作原理
- T12和T13、ADC模块联动，实现对执行器的工作电流采样（主动规避执行动作初期信号扰动）的工作原理
- 电控燃油喷射半桥驱动电路工作原理（部分内容将在后续章节继续介绍）
- QSPI通讯的工作原理，SCLK、MTSR、MRST、SLSO等信号的全称及作用，理解用Comparessed View描述QSPI通讯过程
- BACON寄存器LAST的含义及作用
- BACON寄存器与ECONz寄存器的功能差异?
- 四种QSPI模式的特点及举例

本章结束，谢谢！

Thank you！